RAILWAY RESERVATION SYSTEM

MINI PROJECT REPORT

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ABSTRACT

The **Railway Reservation System** is a web-based application developed to simplify and automate the process of booking train tickets. The system is designed using a frontend interface integrated with a MySQL database managed through phpMyAdmin and hosted locally using the XAMPP server environment. This application enables users to search for available trains, check schedules, and reserve tickets efficiently. It provides functionalities such as user registration, login authentication, seat availability checking, ticket booking, and cancellation. The system ensures secure storage and management of passenger and booking data using a structured database. On the administrative side, the system allows management of train details, routes, schedules, and user records. By replacing the traditional manual booking system, this project minimizes errors, reduces time consumption, and enhances user convenience. The Railway Reservation System demonstrates the effective use of web technologies and database management systems to build a reliable, scalable, and user-friendly application for real-time reservation services. The system is designed with a user-friendly interface to ensure easy navigation for both users and administrators. It supports real-time data processing, allowing instant updates on seat availability and booking status. Data integrity and security are maintained through proper validation and authentication mechanisms. The project can be further enhanced by integrating online payment gateways and live train tracking features. Overall, the system improves efficiency, accuracy, and transparency in railway ticket reservation processes.

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INTRODUCTION

In today's fast-paced world, transportation plays a vital role in connecting people and places. Among various modes of transport, railways are one of the most widely used and economical systems. With the increasing number of passengers, managing ticket reservations manually has become difficult, time-consuming, and prone to errors. To overcome these challenges, computerized reservation systems have been introduced. The **Railway Reservation System** project is developed to provide an efficient, reliable, and user-friendly solution for booking train tickets.

The main objective of this project is to automate the process of railway ticket booking and management. This system allows users to perform various operations such as searching for trains, checking seat availability, booking tickets, and cancelling reservations. It reduces the workload of manual systems and ensures accuracy in maintaining passenger records. The system also helps in minimizing human errors and provides a faster way of handling reservations.

This project is developed using web technologies and a database management system. The frontend of the application is designed to provide an interactive interface for users, while the backend is managed using a MySQL database. The database is created and maintained using **phpMyAdmin**, which is a powerful tool for handling database operations. The application is hosted locally using the **XAMPP server**, which provides an environment to run PHP and MySQL together.

XAMPP is an open-source software package that includes Apache server, MySQL database, PHP, and Perl. It acts as a local server that allows developers to run web applications on their personal computers without the need for an

internet connection. In this project, XAMPP is used to host the Railway Reservation System and manage server-side operations. The Apache server in XAMPP handles HTTP requests, while MySQL stores and retrieves data efficiently.

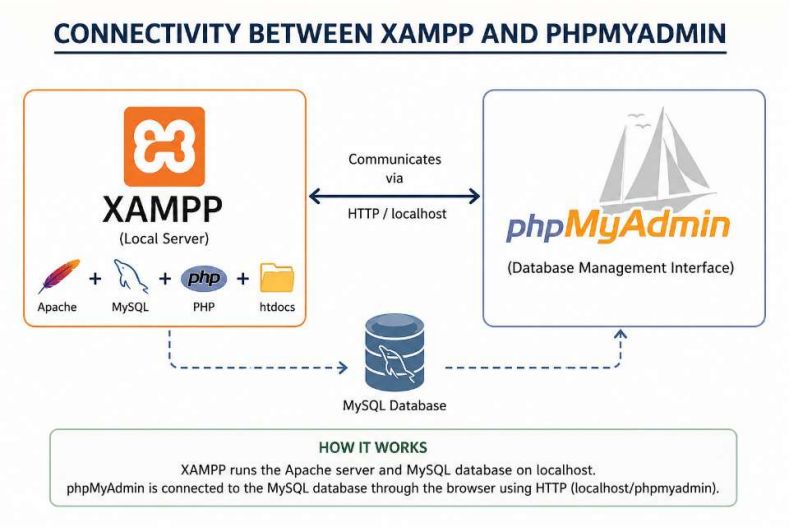
phpMyAdmin is a web-based interface used for managing MySQL databases. It provides an easy way to create, modify, and delete databases, tables, and records. In this project, phpMyAdmin is used to design the database structure for storing information such as user details, train schedules, ticket bookings, and cancellations. It simplifies database management and allows developers to execute SQL queries without writing complex commands manually. The connectivity between the frontend and backend is established using PHP. PHP acts as a server-side scripting language that communicates with the MySQL database.
When a user performs an action such as booking a ticket, the request is sent to the server through PHP scripts. These scripts process the request, interact with the database, and return the required output to the user interface. This ensures smooth communication between the user and the system.

The system follows a structured approach to manage data efficiently. It includes modules such as user registration and login, train information management, ticket booking, and ticket cancellation. Each module is connected to the database and performs specific functions. The login module ensures that only authorized users can access the system, while the booking module handles reservation details and seat allocation.

One of the important features of this system is real-time data handling. When a ticket is booked or cancelled, the database is updated instantly. This helps in maintaining accurate information about seat availability. The system also ensures

data security by validating user inputs and restricting unauthorized access.

In addition to user functionalities, the system provides administrative features. The admin can add new trains, update schedules, and manage user records. This helps in maintaining the system effectively and ensures smooth operation. The centralized database makes it easier to store and retrieve large amounts of data without redundancy.



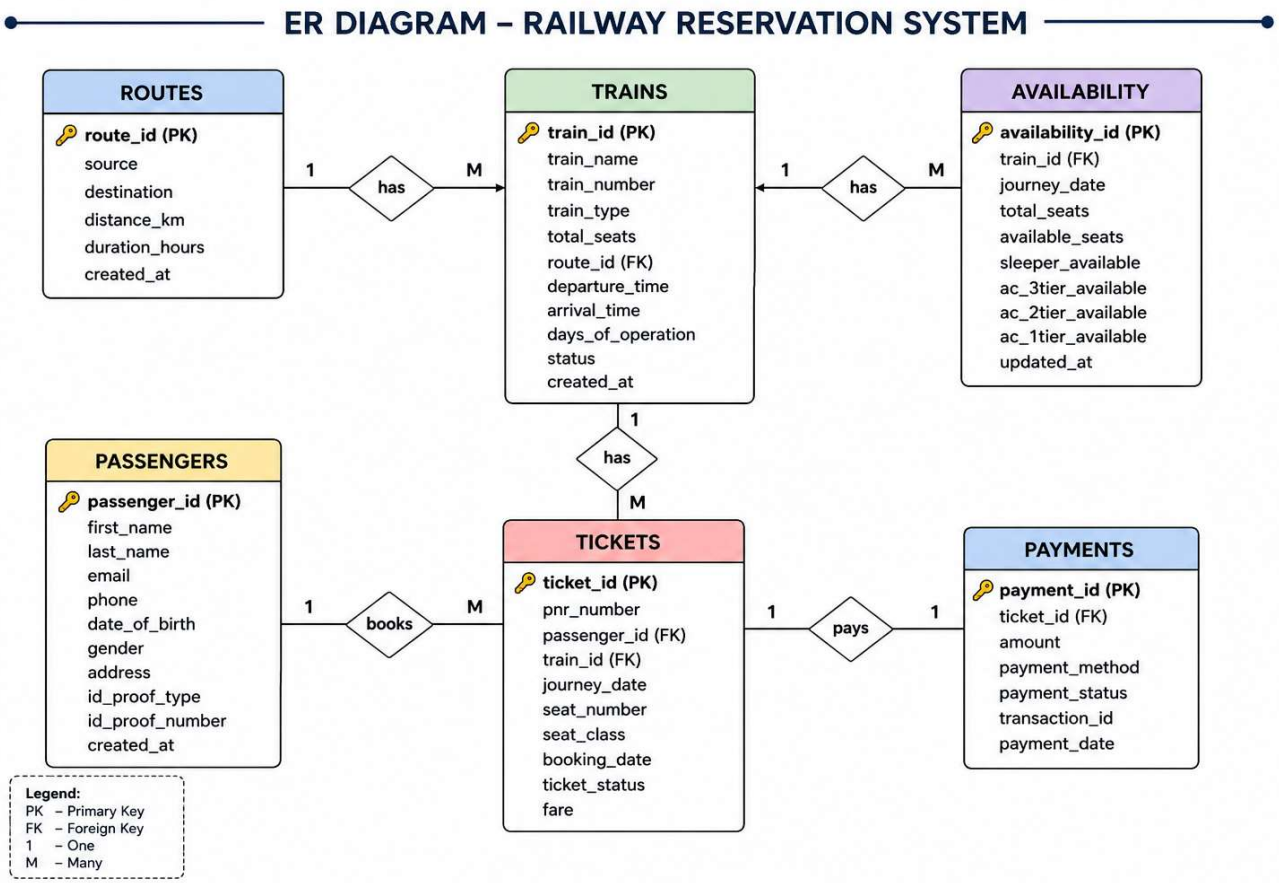
The Railway Reservation System project demonstrates the practical implementation of database management and web development concepts. It shows how frontend interfaces, server-side scripting, and databases work together to create a complete application. The use of XAMPP and phpMyAdmin makes the development process simple and efficient.

In conclusion, this project provides a modern solution to the traditional railway reservation process. It improves efficiency, reduces manual effort, and enhances user experience. The system can be further improved by adding features such as online payment integration, mobile application support, and real-time train tracking. Overall, the Railway Reservation System is a useful and scalable application that meets the needs of modern transportation systems.

DATABASE SCHEMA

```
DATABASE: railway_reservation_system

ROUTES(
   route_id INT PRIMARY KEY,
   source VARCHAR(100),
   destination VARCHAR(100),
   distance_km DECIMAL(10,2),
   duration_hours DECIMAL(5,2)
)

TRAINS(
   train_id INT PRIMARY KEY,
   train_name VARCHAR(100),
   train_number VARCHAR(20) UNIQUE,
   train_type ENUM('Express','Superfast','Passenger','Shatabdi','Rajdhani'),
   total_seats INT,
   route_id INT,
   departure_time TIME,
   arrival_time TIME,
   days_of_operation VARCHAR(50),
   status ENUM('Active','Inactive','Maintenance'),

   FOREIGN KEY (route_id) REFERENCES ROUTES(route_id)
)

PASSENGERS(
   passenger_id INT PRIMARY KEY,
   first_name VARCHAR(50),
   last_name VARCHAR(50),
   email VARCHAR(100) UNIQUE,
   phone VARCHAR(15),
   date_of_birth DATE,
   gender ENUM('Male','Female','Other'),
   address TEXT,
   id_proof_type ENUM('Aadhar','PAN','Passport','Driving License'),
   id_proof_number VARCHAR(50)
)

TICKETS(
   ticket_id INT PRIMARY KEY,
```

```
   pnr_number VARCHAR(10) UNIQUE,
   passenger_id INT,
   train_id INT,
   journey_date DATE,
   seat_number VARCHAR(10),
   seat_class ENUM('Sleeper','3AC','2AC','1AC'),
   booking_date TIMESTAMP,
   ticket_status ENUM('Confirmed','Waiting','Cancelled','RAC'),
   fare DECIMAL(10,2),

   FOREIGN KEY (passenger_id) REFERENCES
PASSENGERS(passenger_id),
   FOREIGN KEY (train_id) REFERENCES TRAINS(train_id)
)

PAYMENTS(
   payment_id INT PRIMARY KEY,
   ticket_id INT,
   amount DECIMAL(10,2),
   payment_method ENUM('Credit Card','Debit Card','UPI','Net
Banking','Wallet'),
   payment_status ENUM('Success','Pending','Failed','Refunded'),
   transaction_id VARCHAR(100) UNIQUE,
   payment_date TIMESTAMP,

   FOREIGN KEY (ticket_id) REFERENCES TICKETS(ticket_id)
)

AVAILABILITY(
   availability_id INT PRIMARY KEY,
   train_id INT,
   journey_date DATE,
   total_seats INT,
   available_seats INT,
   sleeper_available INT,
   ac_3tier_available INT,
   ac_2tier_available INT,
   ac_1tier_available INT,

   FOREIGN KEY (train_id) REFERENCES TRAINS(train_id)

)
```


ER DIAGRAM – RAILWAY RESERVATION SYSTEM
ROUTES
route_id (PK)
source
destination
distance_km
duration_hours
created_at
1
has
M
TRAINS
train_id (PK)
train_name
train_number
train_type
total_seats
route_id (FK)
departure_time
arrival_time
days_of_operation
status
created_at
1
has
M
AVAILABILITY
availability_id (PK)
train_id (FK)
journey_date
total_seats
available_seats
sleeper_available
ac_3tier_available
ac_2tier_available
ac_1tier_available
updated_at
1
has
M
PASSENGERS
passenger_id (PK)
first_name
last_name
email
phone
date_of_birth
gender
address
id_proof_type
id_proof_number
created_at
1
books
M
TICKETS
ticket_id (PK)
pnr_number
passenger_id (FK)
train_id (FK)
journey_date
seat_number
seat_class
booking_date
ticket_status
fare
1
pays
1
PAYMENTS
payment_id (PK)
ticket_id (FK)
amount
payment_method
payment_status
transaction_id
payment_date
Legend:
PK – Primary Key
FK – Foreign Key
1 – One
M – Many

SETUP AND CODING

In the Railway Reservation System project, the connectivity between the frontend and backend is established using a local server environment. The frontend is designed using web technologies such as HTML, CSS, and JavaScript, while the backend is handled using PHP. The database is created and managed using **MySQL through phpMyAdmin**. To integrate all these components, XAMPP is used as a local development server that enables smooth communication between the user interface and the database. This connectivity setup allows admin to perform operations such as viewing the booked tickets, viewing train details, checking seat availability, and making payments, with all data being stored and retrieved dynamically from the database.

XAMPP Server Setup

XAMPP is a cross-platform web server solution that provides an easy way to run **Apache** and **MySQL** on a local machine. It includes Apache server, MySQL database, PHP interpreter, and phpMyAdmin tool. After installing XAMPP, the Apache and MySQL modules are started using the XAMPP Control Panel. Apache is responsible for executing PHP scripts and serving web pages, while MySQL manages the database operations. The project files are stored in the **htdocs** directory inside the XAMPP installation folder, which acts as the root directory for hosting local websites. By running the project through **http://localhost/**, the frontend pages are accessed, and the backend scripts are executed on the server.

Database Creation Using phpMyAdmin

phpMyAdmin is a web-based tool used to manage MySQL databases. It provides a graphical interface to create databases, tables, and relationships without writing complex SQL commands. In this project, a database named railway_reservation_system is created. Inside this database, multiple tables are defined such as trains, routes, passengers, tickets, payments, and availability. Each table contains specific attributes relevant to its purpose. For example, the trains table stores train details, the passengers table stores user information, and the tickets table acts as the central table linking passengers and trains. Primary keys are defined to uniquely identify each record, and foreign keys are used to establish relationships between tables, ensuring referential integrity.

Backend Connectivity Using PHP

PHP is used as the server-side scripting language to connect the frontend with the database. It acts as a bridge that processes user requests and interacts with the MySQL database. The connection to the database is established using the **mysqli_connect()** function, where the hostname, username, password, and database name are specified. Once the connection is successful, PHP scripts execute SQL queries such as INSERT, SELECT, UPDATE, and DELETE.

For example, when a user books a ticket, the form data is sent to a PHP file, which inserts the data into the tickets table. Similarly, when retrieving train or booking details, PHP executes SELECT queries and displays the results on the frontend. PHP also ensures data validation and error handling, making the system reliable and secure.

Frontend Interaction

The frontend is developed using HTML for structure, CSS for styling, and JavaScript for interactivity. It provides a user-friendly interface where users can enter details such as passenger information, select trains, and book tickets. Forms are used to collect user input, and the data is sent to PHP scripts using HTTP methods like **GET and POST**. The frontend does not directly interact with the database; instead, it communicates through PHP, ensuring proper **data handling and security**. Dynamic content such as available trains, ticket details, and booking confirmations are displayed by fetching data from the backend.

Data Flow Process

The connectivity between frontend and backend follows a structured flow. When a user interacts with the interface, the request is sent to the server where PHP processes it. PHP communicates with the MySQL database to perform the required operation and then sends the response back to the frontend.

The flow can be summarized as follows:

User Interface → PHP Script (Server) → MySQL Database → PHP Script → User Interface

This process ensures that all user actions are processed efficiently and the results are displayed in real time.

Advantages of the Setup

This connectivity setup offers several advantages. It is easy to implement and cost-effective as it uses open-source tools. XAMPP provides a complete development environment, making it suitable for beginners. The use of PHP and MySQL ensures fast processing and efficient data management. Additionally, phpMyAdmin simplifies database handling through its graphical interface.

Coding

railway_reservation_system.sql

```
-- Railway Reservation System Database Schema
-- Create Database
CREATE DATABASE IF NOT EXISTS railway_reservation_system;
USE railway_reservation_system;

-- Drop tables if they exist (for fresh installation)
DROP TABLE IF EXISTS PAYMENT;
DROP TABLE IF EXISTS TICKET;
DROP TABLE IF EXISTS SEAT_AVAILABILITY;
DROP TABLE IF EXISTS PASSENGER;
DROP TABLE IF EXISTS TRAIN;
DROP TABLE IF EXISTS ROUTE;

-- 1. ROUTE Table
CREATE TABLE ROUTE (
    route_id INT PRIMARY KEY AUTO_INCREMENT,
    source VARCHAR(100) NOT NULL,
    destination VARCHAR(100) NOT NULL,
    distance_km DECIMAL(10, 2) NOT NULL,
    duration_hours DECIMAL(5, 2) NOT NULL,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

-- 2. TRAIN Table
CREATE TABLE TRAIN (
    train_id INT PRIMARY KEY AUTO_INCREMENT,
    train_name VARCHAR(100) NOT NULL,
    train_number VARCHAR(20) UNIQUE NOT NULL,
    train_type ENUM('Express', 'Superfast', 'Passenger', 'Shatabdi',
'Rajdhani') NOT NULL,
    total_seats INT NOT NULL,
    route_id INT NOT NULL,
    departure_time TIME NOT NULL,
    arrival_time TIME NOT NULL,
    days_of_operation VARCHAR(50) DEFAULT 'All Days',
    status   ENUM('Active',   'Inactive',   'Maintenance')   DEFAULT
'Active',
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    FOREIGN  KEY  (route_id)  REFERENCES  ROUTE(route_id)  ON  DELETE
CASCADE ON UPDATE CASCADE
);

-- 3. PASSENGER Table
CREATE TABLE PASSENGER (
```

```
    passenger_id INT PRIMARY KEY AUTO_INCREMENT,
    first_name VARCHAR(50) NOT NULL,
    last_name VARCHAR(50) NOT NULL,
    email VARCHAR(100) UNIQUE NOT NULL,
    phone VARCHAR(15) NOT NULL,
    date_of_birth DATE NOT NULL,
    gender ENUM('Male', 'Female', 'Other') NOT NULL,
    address TEXT,
    id_proof_type   ENUM('Aadhar',   'PAN',   'Passport',   'Driving
License') NOT NULL,
    id_proof_number VARCHAR(50) NOT NULL,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

-- 4. SEAT_AVAILABILITY Table
CREATE TABLE SEAT_AVAILABILITY (
    availability_id INT PRIMARY KEY AUTO_INCREMENT,
    train_id INT NOT NULL,
    journey_date DATE NOT NULL,
    total_seats INT NOT NULL,
    available_seats INT NOT NULL,
    sleeper_available INT DEFAULT 0,
    ac_3tier_available INT DEFAULT 0,
    ac_2tier_available INT DEFAULT 0,
    ac_1tier_available INT DEFAULT 0,
    updated_at   TIMESTAMP   DEFAULT   CURRENT_TIMESTAMP   ON   UPDATE
CURRENT_TIMESTAMP,
    FOREIGN  KEY  (train_id)  REFERENCES  TRAIN(train_id)  ON  DELETE
CASCADE ON UPDATE CASCADE,
    UNIQUE KEY unique_train_date (train_id, journey_date)
);

-- 5. TICKET Table
CREATE TABLE TICKET (
    ticket_id INT PRIMARY KEY AUTO_INCREMENT,
    pnr_number VARCHAR(10) UNIQUE NOT NULL,
    passenger_id INT NOT NULL,
    train_id INT NOT NULL,
    journey_date DATE NOT NULL,
    seat_number VARCHAR(10) NOT NULL,
    seat_class ENUM('Sleeper', '3AC', '2AC', '1AC') NOT NULL,
    booking_date TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    ticket_status  ENUM('Confirmed',  'Waiting',  'Cancelled',  'RAC')
DEFAULT 'Confirmed',
    fare DECIMAL(10, 2) NOT NULL,
    FOREIGN KEY (passenger_id) REFERENCES PASSENGER(passenger_id) ON
DELETE CASCADE ON UPDATE CASCADE,
    FOREIGN  KEY  (train_id)  REFERENCES  TRAIN(train_id)  ON  DELETE
CASCADE ON UPDATE CASCADE
```

```
);

-- 6. PAYMENT Table
CREATE TABLE PAYMENT (
    payment_id INT PRIMARY KEY AUTO_INCREMENT,
    ticket_id INT NOT NULL,
    amount DECIMAL(10, 2) NOT NULL,
    payment_method ENUM('Credit Card', 'Debit Card', 'UPI', 'Net
Banking', 'Wallet') NOT NULL,
    payment_status ENUM('Success', 'Pending', 'Failed', 'Refunded')
DEFAULT 'Pending',
    transaction_id VARCHAR(100) UNIQUE NOT NULL,
    payment_date TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    FOREIGN KEY (ticket_id) REFERENCES TICKET(ticket_id) ON DELETE
CASCADE ON UPDATE CASCADE
);

-- Create Indexes for Better Performance
CREATE INDEX idx_train_route ON TRAIN(route_id);
CREATE INDEX idx_ticket_passenger ON TICKET(passenger_id);
CREATE INDEX idx_ticket_train ON TICKET(train_id);
CREATE INDEX idx_ticket_pnr ON TICKET(pnr_number);
CREATE INDEX idx_payment_ticket ON PAYMENT(ticket_id);
CREATE INDEX idx_seat_train_date ON SEAT_AVAILABILITY(train_id,
journey_date);

-- Insert Sample Data for ROUTE
INSERT INTO ROUTE (source, destination, distance_km, duration_hours)
VALUES
('Chennai Central', 'Mumbai Central', 1338.50, 24.00),
('Delhi Junction', 'Kolkata Howrah', 1441.00, 17.30),
('Bangalore City', 'Chennai Central', 362.00, 6.00),
('Mumbai Central', 'Delhi Junction', 1384.00, 16.00),
('Hyderabad Deccan', 'Bangalore City', 574.00, 10.00),
('Chennai Central', 'Coimbatore Junction', 497.00, 7.30),
('Delhi Junction', 'Jaipur Junction', 308.00, 5.00),
('Mumbai Central', 'Pune Junction', 192.00, 3.30);

-- Insert Sample Data for TRAIN
INSERT INTO TRAIN (train_name, train_number, train_type, total_seats,
route_id, departure_time, arrival_time, days_of_operation, status)
VALUES
('Chennai Mumbai Express', '12164', 'Express', 1200, 1, '06:00:00',
'06:00:00', 'All Days', 'Active'),
('Rajdhani Express', '12301', 'Rajdhani', 800, 2, '16:55:00',
'10:25:00', 'All Days', 'Active'),
('Shatabdi Express', '12028', 'Shatabdi', 600, 3, '06:00:00',
'12:00:00', 'All Days', 'Active'),
('Duronto Express', '12263', 'Superfast', 1000, 4, '21:15:00',
```

```
'13:15:00', 'All Days', 'Active'),
('Intercity Express', '12785', 'Express', 700, 5, '05:30:00',
'15:30:00', 'All Days', 'Active'),
('Nilgiri Express', '12671', 'Express', 900, 6, '20:30:00',
'04:00:00', 'All Days', 'Active'),
('Jaipur Express', '12413', 'Express', 850, 7, '07:00:00', '12:00:00',
'All Days', 'Active'),
('Deccan Queen', '12123', 'Express', 750, 8, '07:10:00', '10:40:00',
'All Days', 'Active');

-- Insert Sample Data for PASSENGER
INSERT INTO PASSENGER (first_name, last_name, email, phone,
date_of_birth, gender, address, id_proof_type, id_proof_number)
VALUES
('Rajesh', 'Kumar', 'rajesh.kumar@email.com', '9876543210', '1985-03-
15', 'Male', '123 MG Road, Chennai', 'Aadhar', '1234-5678-9012'),
('Priya', 'Sharma', 'priya.sharma@email.com', '9876543211', '1990-07-
22', 'Female', '456 Park Street, Mumbai', 'PAN', 'ABCDE1234F'),
('Amit', 'Patel', 'amit.patel@email.com', '9876543212', '1988-11-10',
'Male', '789 Brigade Road, Bangalore', 'Passport', 'M1234567'),
('Sneha', 'Reddy', 'sneha.reddy@email.com', '9876543213', '1992-05-
18', 'Female', '321 Connaught Place, Delhi', 'Driving License', 'DL-
1234567890'),
('Vijay', 'Mehta', 'vijay.mehta@email.com', '9876543214', '1987-09-
25', 'Male', '654 MG Road, Pune', 'Aadhar', '9876-5432-1098');

-- Insert Sample Data for SEAT_AVAILABILITY (for next 7 days)
INSERT INTO SEAT_AVAILABILITY (train_id, journey_date, total_seats,
available_seats, sleeper_available, ac_3tier_available,
ac_2tier_available, ac_1tier_available) VALUES
(1, DATE_ADD(CURDATE(), INTERVAL 1 DAY), 1200, 845, 450, 250, 100,
45),
(1, DATE_ADD(CURDATE(), INTERVAL 2 DAY), 1200, 920, 500, 280, 100,
40),
(2, DATE_ADD(CURDATE(), INTERVAL 1 DAY), 800, 567, 300, 180, 60, 27),
(2, DATE_ADD(CURDATE(), INTERVAL 3 DAY), 800, 650, 350, 200, 70, 30),
(3, DATE_ADD(CURDATE(), INTERVAL 1 DAY), 600, 425, 250, 120, 40, 15),
(4, DATE_ADD(CURDATE(), INTERVAL 2 DAY), 1000, 780, 400, 250, 90, 40),
(5, DATE_ADD(CURDATE(), INTERVAL 1 DAY), 700, 530, 300, 150, 60, 20),
(6, DATE_ADD(CURDATE(), INTERVAL 2 DAY), 900, 680, 380, 200, 70, 30);

-- Insert Sample Data for TICKET
INSERT INTO TICKET (pnr_number, passenger_id, train_id, journey_date,
seat_number, seat_class, ticket_status, fare) VALUES
('PNR1234567', 1, 1, DATE_ADD(CURDATE(), INTERVAL 1 DAY), 'S5-42',
'Sleeper', 'Confirmed', 850.00),
('PNR2345678', 2, 2, DATE_ADD(CURDATE(), INTERVAL 1 DAY), 'A1-15',
'2AC', 'Confirmed', 2500.00),
('PNR3456789', 3, 3, DATE_ADD(CURDATE(), INTERVAL 1 DAY), 'B3-28',
```

```
'3AC', 'Confirmed', 1200.00),
('PNR4567890', 4, 4, DATE_ADD(CURDATE(), INTERVAL 2 DAY), 'S7-56',
'Sleeper', 'Confirmed', 920.00),
('PNR5678901', 5, 5, DATE_ADD(CURDATE(), INTERVAL 1 DAY), 'A2-12',
'1AC', 'Confirmed', 3500.00);

-- Insert Sample Data for PAYMENT
INSERT   INTO   PAYMENT   (ticket_id,   amount,   payment_method,
payment_status, transaction_id) VALUES
(1, 850.00, 'UPI', 'Success', 'TXN1001234567890'),
(2, 2500.00, 'Credit Card', 'Success', 'TXN1001234567891'),
(3, 1200.00, 'Debit Card', 'Success', 'TXN1001234567892'),
(4, 920.00, 'Net Banking', 'Success', 'TXN1001234567893'),
(5, 3500.00, 'UPI', 'Success', 'TXN1001234567894');

-- Create Views for Common Queries
CREATE OR REPLACE VIEW ticket_details AS
SELECT
    t.ticket_id,
    t.pnr_number,
    CONCAT(p.first_name, ' ', p.last_name) AS passenger_name,
    p.email,
    p.phone,
    tr.train_name,
    tr.train_number,
    r.source,
    r.destination,
    t.journey_date,
    t.seat_number,
    t.seat_class,
    t.ticket_status,
    t.fare,
    pay.payment_status
FROM TICKET t
JOIN PASSENGER p ON t.passenger_id = p.passenger_id
JOIN TRAIN tr ON t.train_id = tr.train_id
JOIN ROUTE r ON tr.route_id = r.route_id
LEFT JOIN PAYMENT pay ON t.ticket_id = pay.ticket_id;

-- Create View for Train Availability
CREATE OR REPLACE VIEW train_availability AS
SELECT
    t.train_id,
    t.train_name,
    t.train_number,
    t.train_type,
    r.source,
    r.destination,
    t.departure_time,
```

```
    t.arrival_time,
    r.distance_km,
    r.duration_hours,
    sa.journey_date,
    sa.available_seats,
    sa.sleeper_available,
    sa.ac_3tier_available,
    sa.ac_2tier_available,
    sa.ac_1tier_available
FROM TRAIN t
JOIN ROUTE r ON t.route_id = r.route_id
LEFT JOIN SEAT_AVAILABILITY sa ON t.train_id = sa.train_id
WHERE t.status = 'Active';

-- Create Stored Procedure for Booking Ticket
DELIMITER //

CREATE PROCEDURE book_ticket(
    IN p_passenger_id INT,
    IN p_train_id INT,
    IN p_journey_date DATE,
    IN p_seat_class VARCHAR(10),
    IN p_fare DECIMAL(10,2),
    OUT p_pnr VARCHAR(10),
    OUT p_ticket_id INT
)
BEGIN
    DECLARE available_count INT;
    DECLARE seat_num VARCHAR(10);

    -- Check seat availability
    SELECT available_seats INTO available_count
    FROM SEAT_AVAILABILITY
    WHERE train_id = p_train_id AND journey_date = p_journey_date;

    IF available_count > 0 THEN
        -- Generate PNR
        SET p_pnr = CONCAT('PNR', LPAD(FLOOR(RAND() * 10000000), 7,
'0'));

        -- Generate seat number
        SET seat_num = CONCAT(
            CASE p_seat_class
                WHEN 'Sleeper' THEN 'S'
                WHEN '3AC' THEN 'B'
                WHEN '2AC' THEN 'A'
                WHEN '1AC' THEN 'F'
            END,
            FLOOR(RAND() * 10) + 1,
```

```
            '-',
            FLOOR(RAND() * 72) + 1
        );

        -- Insert ticket
        INSERT INTO TICKET (pnr_number, passenger_id, train_id,
journey_date, seat_number, seat_class, fare)
        VALUES (p_pnr, p_passenger_id, p_train_id, p_journey_date,
seat_num, p_seat_class, p_fare);

        SET p_ticket_id = LAST_INSERT_ID();

        -- Update seat availability
        UPDATE SEAT_AVAILABILITY
        SET available_seats = available_seats - 1,
            sleeper_available = CASE WHEN p_seat_class = 'Sleeper'
THEN sleeper_available - 1 ELSE sleeper_available END,
            ac_3tier_available = CASE WHEN p_seat_class = '3AC' THEN
ac_3tier_available - 1 ELSE ac_3tier_available END,
            ac_2tier_available = CASE WHEN p_seat_class = '2AC' THEN
ac_2tier_available - 1 ELSE ac_2tier_available END,
            ac_1tier_available = CASE WHEN p_seat_class = '1AC' THEN
ac_1tier_available - 1 ELSE ac_1tier_available END
        WHERE train_id = p_train_id AND journey_date = p_journey_date;
    ELSE
        SET p_pnr = NULL;
        SET p_ticket_id = NULL;
    END IF;
END //

DELIMITER ;

-- Create Trigger to Update Seat Availability on Ticket Cancellation
DELIMITER //

CREATE TRIGGER after_ticket_cancel
AFTER UPDATE ON TICKET
FOR EACH ROW
BEGIN
    IF NEW.ticket_status = 'Cancelled' AND OLD.ticket_status !=
'Cancelled' THEN
        UPDATE SEAT_AVAILABILITY
        SET available_seats = available_seats + 1,
            sleeper_available = CASE WHEN NEW.seat_class = 'Sleeper'
THEN sleeper_available + 1 ELSE sleeper_available END,
            ac_3tier_available = CASE WHEN NEW.seat_class = '3AC' THEN
ac_3tier_available + 1 ELSE ac_3tier_available END,
            ac_2tier_available = CASE WHEN NEW.seat_class = '2AC' THEN
ac_2tier_available + 1 ELSE ac_2tier_available END,
```

```
            ac_1tier_available = CASE WHEN NEW.seat_class = '1AC' THEN
ac_1tier_available + 1 ELSE ac_1tier_available END
        WHERE   train_id   =   NEW.train_id   AND   journey_date   =
NEW.journey_date;
    END IF;
END //

DELIMITER ;

-- Grant Privileges (adjust username and password as needed)
--   GRANT   ALL   PRIVILEGES   ON   railway_reservation_system.*   TO
'railway_user'@'localhost' IDENTIFIED BY 'railway_pass';
-- FLUSH PRIVILEGES;

SELECT 'Database schema created successfully!' AS status;
```

After the setup, create the database using the SQL Queries on phpMyAdmin.

Frontend

index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta   name="viewport"   content="width=device-width,   initial-
scale=1.0">
    <title>Railway Reservation System - Admin Dashboard</title>
    <link
href="https://fonts.googleapis.com/css2?family=Unbounded:wght@300;40
0;600;700&family=Work+Sans:wght@300;400;500;600&display=swap"
rel="stylesheet">
    <link                                          rel="stylesheet"
href="https://cdnjs.cloudflare.com/ajax/libs/font-
awesome/6.4.0/css/all.min.css">
    <style>
        :root {
            --primary: #0f172a;
            --primary-light: #1e293b;
            --accent: #3b82f6;
            --accent-hover: #2563eb;
            --success: #10b981;
            --warning: #f59e0b;
            --danger: #ef4444;
            --text-primary: #f8fafc;
            --text-secondary: #94a3b8;
            --bg-dark: #020617;
```

```
            --bg-card: #0f172a;
            --border: #334155;
            --shadow: rgba(0, 0, 0, 0.3);
        }

        * {
            margin: 0;
            padding: 0;
            box-sizing: border-box;
        }

        body {
            font-family: 'Work Sans', sans-serif;
            background: linear-gradient(135deg, #020617 0%, #0f172a
100%);
            color: var(--text-primary);
            min-height: 100vh;
            overflow-x: hidden;
        }

        /* Animated Background */
        body::before {
            content: '';
            position: fixed;
            top: 0;
            left: 0;
            width: 100%;
            height: 100%;
            background:
                radial-gradient(circle at 20% 30%, rgba(59, 130, 246,
0.1) 0%, transparent 50%),
                radial-gradient(circle at 80% 70%, rgba(16, 185, 129,
0.08) 0%, transparent 50%),
                radial-gradient(circle at 50% 50%, rgba(245, 158, 11,
0.05) 0%, transparent 50%);
            pointer-events: none;
            z-index: 0;
        }

        .container {
            position: relative;
            z-index: 1;
            max-width: 1600px;
            margin: 0 auto;
            padding: 2rem;
        }

        /* Header */
        header {
```

```
            background: rgba(15, 23, 42, 0.6);
            backdrop-filter: blur(20px);
            border: 1px solid var(--border);
            border-radius: 20px;
            padding: 2rem;
            margin-bottom: 2rem;
            box-shadow: 0 20px 60px var(--shadow);
            animation: slideDown 0.6s ease-out;
        }

        @keyframes slideDown {
            from {
                opacity: 0;
                transform: translateY(-30px);
            }
            to {
                opacity: 1;
                transform: translateY(0);
            }
        }

        .header-content {
            display: flex;
            justify-content: space-between;
            align-items: center;
            flex-wrap: wrap;
            gap: 1.5rem;
        }

        .logo {
            display: flex;
            align-items: center;
            gap: 1rem;
        }

        .logo-icon {
            width: 50px;
            height: 50px;
            background:  linear-gradient(135deg,  var(--accent)  0%,
#8b5cf6 100%);
            border-radius: 12px;
            display: flex;
            align-items: center;
            justify-content: center;
            font-size: 1.5rem;
            box-shadow: 0 8px 24px rgba(59, 130, 246, 0.3);
        }

        h1 {
```

```
            font-family: 'Unbounded', sans-serif;
            font-size: 2rem;
            font-weight: 700;
            background:  linear-gradient(135deg,  var(--text-primary)
0%, var(--accent) 100%);
            -webkit-background-clip: text;
            -webkit-text-fill-color: transparent;
            background-clip: text;
        }

        .header-stats {
            display: flex;
            gap: 2rem;
            animation: fadeIn 0.8s ease-out 0.3s both;
        }

        @keyframes fadeIn {
            from { opacity: 0; }
            to { opacity: 1; }
        }

        .stat-item {
            text-align: center;
        }

        .stat-value {
            font-family: 'Unbounded', sans-serif;
            font-size: 1.8rem;
            font-weight: 700;
            color: var(--accent);
        }

        .stat-label {
            font-size: 0.85rem;
            color: var(--text-secondary);
            margin-top: 0.25rem;
        }

        /* Navigation Tabs */
        .nav-tabs {
            display: flex;
            gap: 0.5rem;
            background: rgba(15, 23, 42, 0.4);
            backdrop-filter: blur(20px);
            padding: 0.5rem;
            border-radius: 16px;
            border: 1px solid var(--border);
            margin-bottom: 2rem;
            overflow-x: auto;
```

```
            animation: slideUp 0.6s ease-out 0.2s both;
        }

        @keyframes slideUp {
            from {
                opacity: 0;
                transform: translateY(20px);
            }
            to {
                opacity: 1;
                transform: translateY(0);
            }
        }

        .nav-tab {
            flex: 1;
            min-width: 120px;
            padding: 1rem 1.5rem;
            background: transparent;
            border: none;
            border-radius: 12px;
            color: var(--text-secondary);
            font-family: 'Work Sans', sans-serif;
            font-size: 0.95rem;
            font-weight: 500;
            cursor: pointer;
            transition: all 0.3s ease;
            display: flex;
            align-items: center;
            justify-content: center;
            gap: 0.5rem;
        }

        .nav-tab:hover {
            background: rgba(59, 130, 246, 0.1);
            color: var(--text-primary);
        }

        .nav-tab.active {
            background:  linear-gradient(135deg,  var(--accent)  0%,
#8b5cf6 100%);
            color: white;
            box-shadow: 0 8px 24px rgba(59, 130, 246, 0.4);
        }

        /* Content Area */
        .content {
            animation: fadeIn 0.6s ease-out 0.4s both;
        }
```

```
.tab-content {
    display: none;
}

.tab-content.active {
    display: block;
    animation: fadeIn 0.4s ease-out;
}

/* Card Styles */
.card {
    background: rgba(15, 23, 42, 0.6);
    backdrop-filter: blur(20px);
    border: 1px solid var(--border);
    border-radius: 20px;
    padding: 2rem;
    box-shadow: 0 20px 60px var(--shadow);
    margin-bottom: 2rem;
}

.card-header {
    display: flex;
    justify-content: space-between;
    align-items: center;
    margin-bottom: 1.5rem;
    padding-bottom: 1rem;
    border-bottom: 1px solid var(--border);
}

.card-title {
    font-family: 'Unbounded', sans-serif;
    font-size: 1.4rem;
    font-weight: 600;
    display: flex;
    align-items: center;
    gap: 0.75rem;
}

.card-title i {
    color: var(--accent);
}

/* Buttons */
.btn {
    padding: 0.75rem 1.5rem;
    border: none;
    border-radius: 12px;
    font-family: 'Work Sans', sans-serif;
```

```
            font-size: 0.95rem;
            font-weight: 500;
            cursor: pointer;
            transition: all 0.3s ease;
            display: inline-flex;
            align-items: center;
            gap: 0.5rem;
            text-decoration: none;
        }

        .btn-primary {
            background:  linear-gradient(135deg,  var(--accent)  0%,
#8b5cf6 100%);
            color: white;
            box-shadow: 0 8px 24px rgba(59, 130, 246, 0.3);
        }

        .btn-primary:hover {
            transform: translateY(-2px);
            box-shadow: 0 12px 32px rgba(59, 130, 246, 0.4);
        }

        .btn-success {
            background: var(--success);
            color: white;
        }

        .btn-success:hover {
            background: #059669;
            transform: translateY(-2px);
        }

        .btn-danger {
            background: var(--danger);
            color: white;
        }

        .btn-danger:hover {
            background: #dc2626;
            transform: translateY(-2px);
        }

        .btn-warning {
            background: var(--warning);
            color: white;
        }

        .btn-warning:hover {
            background: #d97706;
```

```
            transform: translateY(-2px);
        }

        .btn-sm {
            padding: 0.5rem 1rem;
            font-size: 0.85rem;
        }

        /* Forms */
        .form-grid {
            display: grid;
            grid-template-columns:   repeat(auto-fit,   minmax(250px,
1fr));
            gap: 1.5rem;
            margin-bottom: 1.5rem;
        }

        .form-group {
            display: flex;
            flex-direction: column;
            gap: 0.5rem;
        }

        .form-group label {
            font-size: 0.9rem;
            font-weight: 500;
            color: var(--text-secondary);
        }

        .form-control {
            padding: 0.75rem 1rem;
            background: rgba(30, 41, 59, 0.6);
            border: 1px solid var(--border);
            border-radius: 10px;
            color: var(--text-primary);
            font-family: 'Work Sans', sans-serif;
            font-size: 0.95rem;
            transition: all 0.3s ease;
        }

        .form-control:focus {
            outline: none;
            border-color: var(--accent);
            background: rgba(30, 41, 59, 0.8);
            box-shadow: 0 0 0 3px rgba(59, 130, 246, 0.1);
        }

        select.form-control {
            cursor: pointer;
```

```
}

/* Tables */
.table-container {
    overflow-x: auto;
    border-radius: 12px;
    border: 1px solid var(--border);
}

table {
    width: 100%;
    border-collapse: collapse;
    font-size: 0.9rem;
}

thead {
    background: rgba(30, 41, 59, 0.8);
}

th {
    padding: 1rem;
    text-align: left;
    font-weight: 600;
    color: var(--text-primary);
    border-bottom: 2px solid var(--border);
}

td {
    padding: 1rem;
    border-bottom: 1px solid rgba(51, 65, 85, 0.3);
}

tbody tr {
    transition: all 0.3s ease;
}

tbody tr:hover {
    background: rgba(59, 130, 246, 0.05);
}

/* Status Badges */
.badge {
    display: inline-block;
    padding: 0.35rem 0.75rem;
    border-radius: 8px;
    font-size: 0.8rem;
    font-weight: 600;
}
```

```
.badge-success {
    background: rgba(16, 185, 129, 0.2);
    color: var(--success);
}

.badge-warning {
    background: rgba(245, 158, 11, 0.2);
    color: var(--warning);
}

.badge-danger {
    background: rgba(239, 68, 68, 0.2);
    color: var(--danger);
}

.badge-info {
    background: rgba(59, 130, 246, 0.2);
    color: var(--accent);
}

/* Modal */
.modal {
    display: none;
    position: fixed;
    top: 0;
    left: 0;
    width: 100%;
    height: 100%;
    background: rgba(0, 0, 0, 0.8);
    backdrop-filter: blur(10px);
    z-index: 1000;
    overflow-y: auto;
    padding: 2rem;
}

.modal.active {
    display: flex;
    align-items: center;
    justify-content: center;
    animation: fadeIn 0.3s ease-out;
}

.modal-content {
    background: var(--bg-card);
    border: 1px solid var(--border);
    border-radius: 20px;
    padding: 2rem;
    max-width: 800px;
    width: 100%;
```

```
    max-height: 90vh;
    overflow-y: auto;
    box-shadow: 0 30px 80px rgba(0, 0, 0, 0.5);
    animation: slideUp 0.4s ease-out;
}

.modal-header {
    display: flex;
    justify-content: space-between;
    align-items: center;
    margin-bottom: 1.5rem;
    padding-bottom: 1rem;
    border-bottom: 1px solid var(--border);
}

.modal-title {
    font-family: 'Unbounded', sans-serif;
    font-size: 1.4rem;
    font-weight: 600;
}

.close-btn {
    background: transparent;
    border: none;
    color: var(--text-secondary);
    font-size: 1.5rem;
    cursor: pointer;
    transition: all 0.3s ease;
    width: 36px;
    height: 36px;
    display: flex;
    align-items: center;
    justify-content: center;
    border-radius: 8px;
}

.close-btn:hover {
    background: rgba(239, 68, 68, 0.2);
    color: var(--danger);
}

/* Loading Spinner */
.loading {
    display: none;
    text-align: center;
    padding: 2rem;
}

.loading.active {
```

```
            display: block;
        }

        .spinner {
            width: 50px;
            height: 50px;
            border: 4px solid rgba(59, 130, 246, 0.2);
            border-top-color: var(--accent);
            border-radius: 50%;
            animation: spin 1s linear infinite;
            margin: 0 auto 1rem;
        }

        @keyframes spin {
            to { transform: rotate(360deg); }
        }

        /* Alert Messages */
        .alert {
            padding: 1rem 1.5rem;
            border-radius: 12px;
            margin-bottom: 1.5rem;
            display: none;
            align-items: center;
            gap: 0.75rem;
            animation: slideDown 0.4s ease-out;
        }

        .alert.active {
            display: flex;
        }

        .alert-success {
            background: rgba(16, 185, 129, 0.2);
            border: 1px solid var(--success);
            color: var(--success);
        }

        .alert-error {
            background: rgba(239, 68, 68, 0.2);
            border: 1px solid var(--danger);
            color: var(--danger);
        }

        /* Dashboard Stats Grid */
        .stats-grid {
            display: grid;
            grid-template-columns:   repeat(auto-fit,   minmax(250px,
1fr));
```

```
            gap: 1.5rem;
            margin-bottom: 2rem;
        }

        .stat-card {
            background: rgba(15, 23, 42, 0.6);
            backdrop-filter: blur(20px);
            border: 1px solid var(--border);
            border-radius: 16px;
            padding: 1.5rem;
            display: flex;
            align-items: center;
            gap: 1.5rem;
            transition: all 0.3s ease;
        }

        .stat-card:hover {
            transform: translateY(-4px);
            box-shadow: 0 16px 48px rgba(0, 0, 0, 0.3);
        }

        .stat-icon {
            width: 60px;
            height: 60px;
            border-radius: 12px;
            display: flex;
            align-items: center;
            justify-content: center;
            font-size: 1.5rem;
        }

        .stat-icon.blue { background: rgba(59, 130, 246, 0.2); color:
var(--accent); }
        .stat-icon.green { background: rgba(16, 185, 129, 0.2); color:
var(--success); }
        .stat-icon.orange { background: rgba(245, 158, 11, 0.2);
color: var(--warning); }
        .stat-icon.purple { background: rgba(139, 92, 246, 0.2);
color: #8b5cf6; }

        .stat-info h3 {
            font-family: 'Unbounded', sans-serif;
            font-size: 1.8rem;
            font-weight: 700;
            margin-bottom: 0.25rem;
        }

        .stat-info p {
            font-size: 0.85rem;
```

```
            color: var(--text-secondary);
        }

        /* Responsive */
        @media (max-width: 768px) {
            .container {
                padding: 1rem;
            }

            h1 {
                font-size: 1.5rem;
            }

            .header-stats {
                flex-direction: column;
                gap: 1rem;
            }

            .nav-tabs {
                flex-direction: column;
            }

            .form-grid {
                grid-template-columns: 1fr;
            }

            table {
                font-size: 0.8rem;
            }

            th, td {
                padding: 0.75rem 0.5rem;
            }
        }

        /* Action Buttons in Table */
        .action-buttons {
            display: flex;
            gap: 0.5rem;
            flex-wrap: wrap;
        }
    </style>
</head>
<body>
    <div class="container">
        <!-- Header -->
        <header>
            <div class="header-content">
                <div class="logo">
```

```
                    <div class="logo-icon">
                        <i class="fas fa-train"></i>
                    </div>
                    <div>
                        <h1>Railway Reservation System</h1>
                        <p style="color: var(--text-secondary); font-
size: 0.9rem; margin-top: 0.25rem;">Admin Dashboard & Management
Portal</p>
                    </div>
                </div>
                <div class="header-stats">
                    <div class="stat-item">
                        <div                      class="stat-value"
id="headerTotalBookings">0</div>
                        <div class="stat-label">Total Bookings</div>
                    </div>
                    <div class="stat-item">
                        <div                      class="stat-value"
id="headerTodayBookings">0</div>
                        <div              class="stat-label">Today's
Bookings</div>
                    </div>
                    <div class="stat-item">
                        <div               class="stat-value">₹<span
id="headerRevenue">0</span></div>
                        <div class="stat-label">Total Revenue</div>
                    </div>
                </div>
            </div>
        </header>

        <!-- Navigation Tabs -->
        <div class="nav-tabs">
            <button              class="nav-tab              active"
onclick="switchTab('dashboard')">
                <i class="fas fa-chart-line"></i> Dashboard
            </button>
            <button class="nav-tab" onclick="switchTab('routes')">
                <i class="fas fa-route"></i> Routes
            </button>
            <button class="nav-tab" onclick="switchTab('trains')">
                <i class="fas fa-train"></i> Trains
            </button>
            <button                                  class="nav-tab"
onclick="switchTab('passengers')">
                <i class="fas fa-users"></i> Passengers
            </button>
            <button class="nav-tab" onclick="switchTab('tickets')">
                <i class="fas fa-ticket-alt"></i> Tickets
```

```
            </button>
            <button class="nav-tab" onclick="switchTab('payments')">
                <i class="fas fa-credit-card"></i> Payments
            </button>
            <button                                  class="nav-tab"
onclick="switchTab('availability')">
                <i class="fas fa-chair"></i> Seat Availability
            </button>
        </div>

        <!-- Alert Messages -->
        <div id="alertSuccess" class="alert alert-success">
            <i class="fas fa-check-circle"></i>
            <span id="successMessage"></span>
        </div>
        <div id="alertError" class="alert alert-error">
            <i class="fas fa-exclamation-circle"></i>
            <span id="errorMessage"></span>
        </div>

        <!-- Content Area -->
        <div class="content">
            <!-- Dashboard Tab -->
            <div id="dashboard" class="tab-content active">
                <div class="stats-grid">
                    <div class="stat-card">
                        <div class="stat-icon blue">
                            <i class="fas fa-route"></i>
                        </div>
                        <div class="stat-info">
                            <h3 id="totalRoutes">0</h3>
                            <p>Total Routes</p>
                        </div>
                    </div>
                    <div class="stat-card">
                        <div class="stat-icon green">
                            <i class="fas fa-train"></i>
                        </div>
                        <div class="stat-info">
                            <h3 id="totalTrains">0</h3>
                            <p>Active Trains</p>
                        </div>
                    </div>
                    <div class="stat-card">
                        <div class="stat-icon orange">
                            <i class="fas fa-users"></i>
                        </div>
                        <div class="stat-info">
                            <h3 id="totalPassengers">0</h3>
```

```
                        <p>Registered Passengers</p>
                    </div>
                </div>
                <div class="stat-card">
                    <div class="stat-icon purple">
                        <i class="fas fa-ticket-alt"></i>
                    </div>
                    <div class="stat-info">
                        <h3 id="totalBookingsCard">0</h3>
                        <p>Total Bookings</p>
                    </div>
                </div>
            </div>

            <div class="card">
                <div class="card-header">
                    <h2 class="card-title">
                        <i class="fas fa-history"></i>
                        Recent Bookings
                    </h2>
                </div>
                <div class="table-container">
                    <table>
                        <thead>
                            <tr>
                                <th>PNR</th>
                                <th>Passenger</th>
                                <th>Train</th>
                                <th>Route</th>
                                <th>Date</th>
                                <th>Seat</th>
                                <th>Class</th>
                                <th>Fare</th>
                                <th>Status</th>
                            </tr>
                        </thead>
                        <tbody id="recentBookingsTable">
                            <tr>
                                <td   colspan="9"   style="text-
align: center; color: var(--text-secondary);">
                                    <div class="loading active">
                                        <div
class="spinner"></div>
                                        Loading           recent
bookings...
                                    </div>
                                </td>
                            </tr>
                        </tbody>
```

```
                        </table>
                    </div>
                </div>
            </div>

            <!-- Routes Tab -->
            <div id="routes" class="tab-content">
                <div class="card">
                    <div class="card-header">
                        <h2 class="card-title">
                            <i class="fas fa-route"></i>
                            Route Management
                        </h2>
                        <button        class="btn        btn-primary"
onclick="openAddRoute()">
                            <i class="fas fa-plus"></i> Add New Route
                        </button>
                    </div>
                    <div class="table-container">
                        <table>
                            <thead>
                                <tr>
                                    <th>ID</th>
                                    <th>Source</th>
                                    <th>Destination</th>
                                    <th>Distance (km)</th>
                                    <th>Duration (hrs)</th>
                                    <th>Actions</th>
                                </tr>
                            </thead>
                            <tbody id="routesTable">
                                <tr>
                                    <td   colspan="6"   style="text-
align: center;">
                                        <div class="loading active">
                                            <div
class="spinner"></div>
                                            Loading routes...
                                        </div>
                                    </td>
                                </tr>
                            </tbody>
                        </table>
                    </div>
                </div>
            </div>

            <!-- Trains Tab -->
            <div id="trains" class="tab-content">
```

```
                <div class="card">
                    <div class="card-header">
                        <h2 class="card-title">
                            <i class="fas fa-train"></i>
                            Train Management
                        </h2>
                        <button        class="btn        btn-primary"
onclick="openAddTrain()">
                            <i class="fas fa-plus"></i> Add New Train
                        </button>
                    </div>
                    <div class="table-container">
                        <table>
                            <thead>
                                <tr>
                                    <th>ID</th>
                                    <th>Train Name</th>
                                    <th>Train Number</th>
                                    <th>Type</th>
                                    <th>Route</th>
                                    <th>Departure</th>
                                    <th>Arrival</th>
                                    <th>Total Seats</th>
                                    <th>Status</th>
                                    <th>Actions</th>
                                </tr>
                            </thead>
                            <tbody id="trainsTable">
                                <tr>
                                    <td   colspan="10"   style="text-
align: center;">
                                        <div class="loading active">
                                            <div
class="spinner"></div>
                                            Loading trains...
                                        </div>
                                    </td>
                                </tr>
                            </tbody>
                        </table>
                    </div>
                </div>
            </div>

            <!-- Passengers Tab -->
            <div id="passengers" class="tab-content">
                <div class="card">
                    <div class="card-header">
                        <h2 class="card-title">
```

```
                            <i class="fas fa-users"></i>
                            Passenger Management
                        </h2>
                        <button        class="btn        btn-primary"
onclick="openAddPassenger()">
                            <i   class="fas   fa-plus"></i>   Add   New
Passenger
                        </button>
                    </div>
                    <div class="table-container">
                        <table>
                            <thead>
                                <tr>
                                    <th>ID</th>
                                    <th>Name</th>
                                    <th>Email</th>
                                    <th>Phone</th>
                                    <th>DOB</th>
                                    <th>Gender</th>
                                    <th>ID Proof</th>
                                    <th>Actions</th>
                                </tr>
                            </thead>
                            <tbody id="passengersTable">
                                <tr>
                                    <td   colspan="8"   style="text-
align: center;">
                                        <div class="loading active">
                                            <div
class="spinner"></div>
                                            Loading passengers...
                                        </div>
                                    </td>
                                </tr>
                            </tbody>
                        </table>
                    </div>
                </div>
            </div>

            <!-- Tickets Tab -->
            <div id="tickets" class="tab-content">
                <div class="card">
                    <div class="card-header">
                        <h2 class="card-title">
                            <i class="fas fa-ticket-alt"></i>
                            Ticket Management
                        </h2>
                        <button        class="btn        btn-primary"
```

```
onclick="openAddTicket()">
                            <i  class="fas  fa-plus"></i>  Book  New
Ticket
                        </button>
                    </div>
                    <div class="table-container">
                        <table>
                            <thead>
                                <tr>
                                    <th>Ticket ID</th>
                                    <th>PNR</th>
                                    <th>Passenger</th>
                                    <th>Train</th>
                                    <th>Route</th>
                                    <th>Journey Date</th>
                                    <th>Seat</th>
                                    <th>Class</th>
                                    <th>Fare</th>
                                    <th>Status</th>
                                    <th>Actions</th>
                                </tr>
                            </thead>
                            <tbody id="ticketsTable">
                                <tr>
                                    <td   colspan="11"   style="text-
align: center;">
                                        <div class="loading active">
                                            <div
class="spinner"></div>
                                            Loading tickets...
                                        </div>
                                    </td>
                                </tr>
                            </tbody>
                        </table>
                    </div>
                </div>
            </div>

            <!-- Payments Tab -->
            <div id="payments" class="tab-content">
                <div class="card">
                    <div class="card-header">
                        <h2 class="card-title">
                            <i class="fas fa-credit-card"></i>
                            Payment Management
                        </h2>
                        <button        class="btn        btn-primary"
onclick="openAddPayment()">
```

```
                            <i  class="fas  fa-plus"></i>  Add  New
Payment
                        </button>
                    </div>
                    <div class="table-container">
                        <table>
                            <thead>
                                <tr>
                                    <th>Payment ID</th>
                                    <th>Transaction ID</th>
                                    <th>PNR Number</th>
                                    <th>Amount</th>
                                    <th>Payment Method</th>
                                    <th>Status</th>
                                    <th>Date</th>
                                    <th>Actions</th>
                                </tr>
                            </thead>
                            <tbody id="paymentsTable">
                                <tr>
                                    <td   colspan="8"   style="text-
align: center;">
                                        <div class="loading active">
                                            <div
class="spinner"></div>
                                            Loading payments...
                                        </div>
                                    </td>
                                </tr>
                            </tbody>
                        </table>
                    </div>
                </div>
            </div>

            <!-- Seat Availability Tab -->
            <div id="availability" class="tab-content">
                <div class="card">
                    <div class="card-header">
                        <h2 class="card-title">
                            <i class="fas fa-chair"></i>
                            Seat Availability Management
                        </h2>
                        <button        class="btn        btn-primary"
onclick="openAddAvailability()">
                            <i  class="fas  fa-plus"></i>  Add  Seat
Availability
                        </button>
                    </div>
```

```
                    <div class="table-container">
                        <table>
                            <thead>
                                <tr>
                                    <th>ID</th>
                                    <th>Train</th>
                                    <th>Journey Date</th>
                                    <th>Total Seats</th>
                                    <th>Available</th>
                                    <th>Sleeper</th>
                                    <th>3AC</th>
                                    <th>2AC</th>
                                    <th>1AC</th>
                                    <th>Actions</th>
                                </tr>
                            </thead>
                            <tbody id="availabilityTable">
                                <tr>
                                    <td   colspan="10"   style="text-
align: center;">
                                        <div class="loading active">
                                            <div
class="spinner"></div>
                                            Loading availability...
                                        </div>
                                    </td>
                                </tr>
                            </tbody>
                        </table>
                    </div>
                </div>
            </div>
        </div>
    </div>

    <!-- Route Modal -->
    <div id="routeModal" class="modal">
        <div class="modal-content">
            <div class="modal-header">
                <h2 class="modal-title" id="routeModalTitle">Add New
Route</h2>
                <button                             class="close-btn"
onclick="closeModal('routeModal')">
                    <i class="fas fa-times"></i>
                </button>
            </div>
            <form id="routeForm">
                <input type="hidden" id="routeId">
                <div class="form-grid">
```

```
                    <div class="form-group">
                        <label              for="routeSource">Source
Station</label>
                        <input      type="text"      id="routeSource"
class="form-control" required>
                    </div>
                    <div class="form-group">
                        <label    for="routeDestination">Destination
Station</label>
                        <input   type="text"   id="routeDestination"
class="form-control" required>
                    </div>
                    <div class="form-group">
                        <label          for="routeDistance">Distance
(km)</label>
                        <input       type="number"       step="0.01"
id="routeDistance" class="form-control" required>
                    </div>
                    <div class="form-group">
                        <label          for="routeDuration">Duration
(hours)</label>
                        <input       type="number"       step="0.01"
id="routeDuration" class="form-control" required>
                    </div>
                </div>
                <div  style="display:  flex;  gap:  1rem;  justify-
content: flex-end; margin-top: 1.5rem;">
                    <button  type="button"  class="btn  btn-danger"
onclick="closeModal('routeModal')">
                        <i class="fas fa-times"></i> Cancel
                    </button>
                    <button type="submit" class="btn btn-success">
                        <i class="fas fa-save"></i> Save Route
                    </button>
                </div>
            </form>
        </div>
    </div>

    <!-- Train Modal -->
    <div id="trainModal" class="modal">
        <div class="modal-content">
            <div class="modal-header">
                <h2 class="modal-title" id="trainModalTitle">Add New
Train</h2>
                <button                             class="close-btn"
onclick="closeModal('trainModal')">
                    <i class="fas fa-times">></i>
                </button>
```

```
            </div>
            <form id="trainForm">
                <input type="hidden" id="trainId">
                <div class="form-grid">
                    <div class="form-group">
                        <label for="trainName">Train Name</label>
                        <input       type="text"       id="trainName"
class="form-control" required>
                    </div>
                    <div class="form-group">
                        <label                for="trainNumber">Train
Number</label>
                        <input      type="text"      id="trainNumber"
class="form-control" required>
                    </div>
                    <div class="form-group">
                        <label for="trainType">Train Type</label>
                        <select  id="trainType"  class="form-control"
required>
                            <option value="">Select Type</option>
                            <option value="Express">Express</option>
                            <option
value="Superfast">Superfast</option>
                            <option
value="Passenger">Passenger</option>
                            <option
value="Shatabdi">Shatabdi</option>
                            <option
value="Rajdhani">Rajdhani</option>
                        </select>
                    </div>
                    <div class="form-group">
                        <label for="trainSeats">Total Seats</label>
                        <input     type="number"     id="trainSeats"
class="form-control" required>
                    </div>
                    <div class="form-group">
                        <label for="trainRoute">Route</label>
                        <select id="trainRoute" class="form-control"
required>
                            <option value="">Select Route</option>
                        </select>
                    </div>
                    <div class="form-group">
                        <label        for="trainDeparture">Departure
Time</label>
                        <input    type="time"    id="trainDeparture"
class="form-control" required>
                    </div>
```

```
                    <div class="form-group">
                        <label            for="trainArrival">Arrival
Time</label>
                        <input     type="time"     id="trainArrival"
class="form-control" required>
                    </div>
                    <div class="form-group">
                        <label        for="trainDays">Days        of
Operation</label>
                        <input       type="text"       id="trainDays"
class="form-control" value="All Days" required>
                    </div>
                    <div class="form-group">
                        <label for="trainStatus">Status</label>
                        <select     id="trainStatus"     class="form-
control" required>
                            <option value="Active">Active</option>
                            <option
value="Inactive">Inactive</option>
                            <option
value="Maintenance">Maintenance</option>
                        </select>
                    </div>
                </div>
                <div  style="display:  flex;  gap:  1rem;  justify-
content: flex-end; margin-top: 1.5rem;">
                    <button  type="button"  class="btn  btn-danger"
onclick="closeModal('trainModal')">
                        <i class="fas fa-times"></i> Cancel
                    </button>
                    <button type="submit" class="btn btn-success">
                        <i class="fas fa-save"></i> Save Train
                    </button>
                </div>
            </form>
        </div>
    </div>

    <!-- Passenger Modal -->
    <div id="passengerModal" class="modal">
        <div class="modal-content">
            <div class="modal-header">
                <h2 class="modal-title" id="passengerModalTitle">Add
New Passenger</h2>
                <button                             class="close-btn"
onclick="closeModal('passengerModal')">
                    <i class="fas fa-times">></i>
                </button>
            </div>
```

```
            <form id="passengerForm">
                <input type="hidden" id="passengerId">
                <div class="form-grid">
                    <div class="form-group">
                        <label        for="passengerFirstName">First
Name</label>
                        <input  type="text"  id="passengerFirstName"
class="form-control" required>
                    </div>
                    <div class="form-group">
                        <label          for="passengerLastName">Last
Name</label>
                        <input   type="text"   id="passengerLastName"
class="form-control" required>
                    </div>
                    <div class="form-group">
                        <label for="passengerEmail">Email</label>
                        <input    type="email"    id="passengerEmail"
class="form-control" required>
                    </div>
                    <div class="form-group">
                        <label for="passengerPhone">Phone</label>
                        <input     type="tel"     id="passengerPhone"
class="form-control" required>
                    </div>
                    <div class="form-group">
                        <label       for="passengerDOB">Date       of
Birth</label>
                        <input     type="date"     id="passengerDOB"
class="form-control" required>
                    </div>
                    <div class="form-group">
                        <label for="passengerGender">Gender</label>
                        <select   id="passengerGender"   class="form-
control" required>
                            <option value="">Select Gender</option>
                            <option value="Male">Male</option>
                            <option value="Female">Female</option>
                            <option value="Other">Other</option>
                        </select>
                    </div>
                    <div class="form-group">
                        <label     for="passengerIDType">ID     Proof
Type</label>
                        <select   id="passengerIDType"   class="form-
control" required>
                            <option value="">Select ID Type</option>
                            <option value="Aadhar">Aadhar</option>
                            <option value="PAN">PAN</option>
```

```
                            <option
value="Passport">Passport</option>
                            <option  value="Driving  License">Driving
License</option>
                        </select>
                    </div>
                    <div class="form-group">
                        <label    for="passengerIDNumber">ID    Proof
Number</label>
                        <input   type="text"   id="passengerIDNumber"
class="form-control" required>
                    </div>
                    <div class="form-group" style="grid-column: 1 / -
1;">
                        <label
for="passengerAddress">Address</label>
                        <textarea id="passengerAddress" class="form-
control" rows="3" required></textarea>
                    </div>
                </div>
                <div  style="display:  flex;  gap:  1rem;  justify-
content: flex-end; margin-top: 1.5rem;">
                    <button  type="button"  class="btn  btn-danger"
onclick="closeModal('passengerModal')">
                        <i class="fas fa-times"></i> Cancel
                    </button>
                    <button type="submit" class="btn btn-success">
                        <i class="fas fa-save"></i> Save Passenger
                    </button>
                </div>
            </form>
        </div>
    </div>

    <!-- Ticket Modal -->
    <div id="ticketModal" class="modal">
        <div class="modal-content">
            <div class="modal-header">
                <h2 class="modal-title">Book New Ticket</h2>
                <button                              class="close-btn"
onclick="closeModal('ticketModal')">
                    <i class="fas fa-times"></i>
                </button>
            </div>
            <form id="ticketForm">
                <div class="form-grid">
                    <div class="form-group">
                        <label
for="ticketPassenger">Passenger</label>
```

```
                        <select   id="ticketPassenger"   class="form-
control" required>
                            <option                  value="">Select
Passenger</option>
                        </select>
                    </div>
                    <div class="form-group">
                        <label for="ticketTrain">Train</label>
                        <select     id="ticketTrain"     class="form-
control" required>
                            <option value="">Select Train</option>
                        </select>
                    </div>
                    <div class="form-group">
                        <label       for="ticketJourneyDate">Journey
Date</label>
                        <input   type="date"   id="ticketJourneyDate"
class="form-control" required>
                    </div>
                    <div class="form-group">
                        <label for="ticketClass">Seat Class</label>
                        <select     id="ticketClass"     class="form-
control" required>
                            <option value="">Select Class</option>
                            <option value="Sleeper">Sleeper</option>
                            <option value="3AC">3AC</option>
                            <option value="2AC">2AC</option>
                            <option value="1AC">1AC</option>
                        </select>
                    </div>
                    <div class="form-group">
                        <label              for="ticketStatus">Ticket
Status</label>
                        <select    id="ticketStatus"    class="form-
control" required>
                            <option
value="Confirmed">Confirmed</option>
                            <option value="Waiting">Waiting</option>
                            <option value="RAC">RAC</option>
                        </select>
                    </div>
                    <div class="form-group">
                        <label for="ticketFare">Fare (₹)</label>
                        <input       type="number"       step="0.01"
id="ticketFare" class="form-control" required>
                    </div>
                </div>
                <div  style="display:  flex;  gap:  1rem;  justify-
content: flex-end; margin-top: 1.5rem;">
```

```
                    <button   type="button"   class="btn   btn-danger"
onclick="closeModal('ticketModal')">
                        <i class="fas fa-times"></i> Cancel
                    </button>
                    <button type="submit" class="btn btn-success">
                        <i class="fas fa-save"></i> Book Ticket
                    </button>
                </div>
            </form>
        </div>
    </div>

    <!-- Payment Modal -->
    <div id="paymentModal" class="modal">
        <div class="modal-content">
            <div class="modal-header">
                <h2 class="modal-title">Add Payment</h2>
                <button                             class="close-btn"
onclick="closeModal('paymentModal')">
                    <i class="fas fa-times"></i>
                </button>
            </div>
            <form id="paymentForm">
                <div class="form-grid">
                    <div class="form-group">
                        <label            for="paymentTicket">Ticket
(PNR)</label>
                        <select    id="paymentTicket"    class="form-
control" required>
                            <option value="">Select Ticket</option>
                        </select>
                    </div>
                    <div class="form-group">
                        <label            for="paymentAmount">Amount
(₹)</label>
                        <input       type="number"       step="0.01"
id="paymentAmount" class="form-control" required>
                    </div>
                    <div class="form-group">
                        <label           for="paymentMethod">Payment
Method</label>
                        <select    id="paymentMethod"    class="form-
control" required>
                            <option value="">Select Method</option>
                            <option    value="Credit    Card">Credit
Card</option>
                            <option     value="Debit     Card">Debit
Card</option>
                            <option value="UPI">UPI</option>
```

```
                            <option      value="Net      Banking">Net
Banking</option>
                            <option value="Wallet">Wallet</option>
                        </select>
                    </div>
                    <div class="form-group">
                        <label           for="paymentStatus">Payment
Status</label>
                        <select    id="paymentStatus"    class="form-
control" required>
                            <option value="Success">Success</option>
                            <option value="Pending">Pending</option>
                            <option value="Failed">Failed</option>
                        </select>
                    </div>
                </div>
                <div  style="display:  flex;  gap:  1rem;  justify-
content: flex-end; margin-top: 1.5rem;">
                    <button  type="button"  class="btn  btn-danger"
onclick="closeModal('paymentModal')">
                        <i class="fas fa-times"></i> Cancel
                    </button>
                    <button type="submit" class="btn btn-success">
                        <i class="fas fa-save"></i> Save Payment
                    </button>
                </div>
            </form>
        </div>
    </div>

    <!-- Availability Modal -->
    <div id="availabilityModal" class="modal">
        <div class="modal-content">
            <div class="modal-header">
                <h2                                class="modal-title"
id="availabilityModalTitle">Add Seat Availability</h2>
                <button                             class="close-btn"
onclick="closeModal('availabilityModal')">
                    <i class="fas fa-times">></i>
                </button>
            </div>
            <form id="availabilityForm">
                <input type="hidden" id="availabilityId">
                <div class="form-grid">
                    <div class="form-group">
                        <label for="availabilityTrain">Train</label>
                        <select  id="availabilityTrain"  class="form-
control" required>
                            <option value="">Select Train</option>
```

```
                        </select>
                    </div>
                    <div class="form-group">
                        <label        for="availabilityDate">Journey
Date</label>
                        <input   type="date"   id="availabilityDate"
class="form-control" required>
                    </div>
                    <div class="form-group">
                        <label         for="availabilityTotal">Total
Seats</label>
                        <input  type="number"  id="availabilityTotal"
class="form-control" required>
                    </div>
                    <div class="form-group">
                        <label  for="availabilityAvailable">Available
Seats</label>
                        <input                          type="number"
id="availabilityAvailable" class="form-control" required>
                    </div>
                    <div class="form-group">
                        <label     for="availabilitySleeper">Sleeper
Available</label>
                        <input                          type="number"
id="availabilitySleeper" class="form-control" required>
                    </div>
                    <div class="form-group">
                        <label             for="availability3AC">3AC
Available</label>
                        <input   type="number"   id="availability3AC"
class="form-control" required>
                    </div>
                    <div class="form-group">
                        <label             for="availability2AC">2AC
Available</label>
                        <input   type="number"   id="availability2AC"
class="form-control" required>
                    </div>
                    <div class="form-group">
                        <label             for="availability1AC">1AC
Available</label>
                        <input   type="number"   id="availability1AC"
class="form-control" required>
                    </div>
                </div>
                <div  style="display:  flex;  gap:  1rem;  justify-
content: flex-end; margin-top: 1.5rem;">
                    <button  type="button"  class="btn  btn-danger"
onclick="closeModal('availabilityModal')">
```

```
                        <i class="fas fa-times"></i> Cancel
                    </button>
                    <button type="submit" class="btn btn-success">
                        <i class="fas fa-save"></i> Save Availability
                    </button>
                </div>
            </form>
        </div>
    </div>

    <script src="app.js"></script>
</body>
</html>
```

Javascript

app.js

```
const API = "http://localhost:8080/Railway/api.php";

// ────────────────────────────────────────────
// UTILITIES
// ────────────────────────────────────────────
function showAlert(type, msg) {
    const  el  =  document.getElementById(type  ===  'success'  ?
'alertSuccess' : 'alertError');
    const  msgEl  =  document.getElementById(type  ===  'success'  ?
'successMessage' : 'errorMessage');
    if (!el || !msgEl) return;
    msgEl.textContent = msg;
    el.classList.add('active');
    setTimeout(() => el.classList.remove('active'), 3500);
}

function openModal(id) {
    document.getElementById(id).classList.add('active');
}

function closeModal(id) {
    document.getElementById(id).classList.remove('active');
}

function badgeHtml(value, map) {
    const cls = map[value] || 'badge-info';
    return `<span class="badge ${cls}">${value}</span>`;
}

const statusBadge = {
    'Active':    'badge-success',    'Inactive':    'badge-danger',
```

```
'Maintenance': 'badge-warning',
    'Confirmed':   'badge-success',   'Cancelled':   'badge-danger',
'Waiting': 'badge-warning', 'RAC': 'badge-info',
    'Success': 'badge-success', 'Failed': 'badge-danger', 'Pending':
'badge-warning', 'Refunded': 'badge-info'
};

// ──────────────────────────────────────────────
// TAB SWITCH
// ──────────────────────────────────────────────
function switchTab(tabId) {
    document.querySelectorAll('.tab-content').forEach(t           =>
t.classList.remove('active'));
    document.querySelectorAll('.nav-tab').forEach(b               =>
b.classList.remove('active'));
    document.getElementById(tabId).classList.add('active');
    event.target.classList.add('active');
}

// ──────────────────────────────────────────────
// DASHBOARD STATS
// ──────────────────────────────────────────────
async function loadStats() {
    try {
        const res  = await fetch(API + '?action=stats');
        const data = await res.json();
        const   set      =   (id,   val)   =>   {   const   el   =
document.getElementById(id); if (el) el.textContent = val; };
        set('headerTotalBookings', data.total_tickets);
        set('headerTodayBookings', data.today_tickets);
        set('headerRevenue',       data.total_revenue);
        set('totalRoutes',         data.total_routes);
        set('totalTrains',         data.total_trains);
        set('totalPassengers',     data.total_passengers);
        set('totalBookingsCard',   data.total_tickets);
    } catch (e) { console.error('Stats error', e); }
}

// ═══════════════════════════════════════════════
//  ROUTES
// ═══════════════════════════════════════════════
let editingRouteId = null;

async function loadRoutes() {
    const res  = await fetch(API + '?table=routes');
    const data = await res.json();
    let html = '';
    data.forEach(r => {
        html += `<tr>
```

```
            <td>${r.route_id}</td>
            <td>${r.source}</td>
            <td>${r.destination}</td>
            <td>${r.distance_km} km</td>
            <td>${r.duration_hours} hrs</td>
            <td><div class="action-buttons">
                <button      class="btn      btn-warning      btn-sm"
onclick="openEditRoute(${r.route_id},'${r.source}','${r.destination}
',${r.distance_km},${r.duration_hours})">
                    <i class="fas fa-edit"></i> Edit
                </button>
                <button      class="btn      btn-danger      btn-sm"
onclick="deleteRoute(${r.route_id})">
                    <i class="fas fa-trash"></i> Delete
                </button>
            </div></td>
        </tr>`;
    });
    document.getElementById('routesTable').innerHTML = html;
}

function openAddRoute() {
    editingRouteId = null;
    document.getElementById('routeModalTitle').textContent = 'Add New
Route';
    document.getElementById('routeForm').reset();
    openModal('routeModal');
}

function openEditRoute(id, src, dst, dist, dur) {
    editingRouteId = id;
    document.getElementById('routeModalTitle').textContent  =  'Edit
Route';
    document.getElementById('routeSource').value      = src;
    document.getElementById('routeDestination').value = dst;
    document.getElementById('routeDistance').value    = dist;
    document.getElementById('routeDuration').value    = dur;
    openModal('routeModal');
}

document.getElementById('routeForm').addEventListener('submit',
async (e) => {
    e.preventDefault();
    const data = {
        source:        document.getElementById('routeSource').value,
        destination:
document.getElementById('routeDestination').value,
        distance_km:
document.getElementById('routeDistance').value,
```

```
        duration_hours:
document.getElementById('routeDuration').value
    };
    if (editingRouteId) {
        data.route_id = editingRouteId;
        await  fetch(API  +  '?action=edit_route',  {  method:  'POST',
body: JSON.stringify(data) });
        showAlert('success', 'Route updated successfully!');
    } else {
        await fetch(API + '?action=add_route', { method: 'POST', body:
JSON.stringify(data) });
        showAlert('success', 'Route added successfully!');
    }
    closeModal('routeModal');
    loadRoutes();
    loadStats();
});

async function deleteRoute(id) {
    if (!confirm('Delete this route? All associated trains will also
be deleted.')) return;
    await fetch(API + '?action=delete_route&id=' + id);
    showAlert('success', 'Route deleted.');
    loadRoutes();
    loadStats();
}

// ═══════════════════════════════════════════
//  TRAINS
// ═══════════════════════════════════════════
let editingTrainId = null;

async function loadTrains() {
    const res  = await fetch(API + '?table=trains');
    const data = await res.json();
    let html = '';
    data.forEach(t => {
        html += `<tr>
            <td>${t.train_id}</td>
            <td>${t.train_name}</td>
            <td>${t.train_number}</td>
            <td>${t.train_type}</td>
            <td>${t.route_id}</td>
            <td>${t.departure_time}</td>
            <td>${t.arrival_time}</td>
            <td>${t.total_seats}</td>
            <td>${badgeHtml(t.status, statusBadge)}</td>
            <td><div class="action-buttons">
                <button      class="btn      btn-warning      btn-sm"
```

```
onclick='openEditTrain(${JSON.stringify(t)})'>
                    <i class="fas fa-edit"></i> Edit
                </button>
                <button      class="btn      btn-danger      btn-sm"
onclick="deleteTrain(${t.train_id})">
                    <i class="fas fa-trash"></i> Delete
                </button>
            </div></td>
        </tr>`;
    });
    document.getElementById('trainsTable').innerHTML = html;
    populateTrainDropdowns(data);
}

function populateTrainDropdowns(trains) {
    ['ticketTrain','availabilityTrain'].forEach(id => {
        const sel = document.getElementById(id);
        if (!sel) return;
        const cur = sel.value;
        sel.innerHTML = '<option value="">Select Train</option>';
        trains.forEach(t => {
            sel.innerHTML                 +=                 `<option
value="${t.train_id}">${t.train_name} (${t.train_number})</option>`;
        });
        sel.value = cur;
    });
}

async function populateRouteDropdown() {
    const res    = await fetch(API + '?table=routes');
    const routes = await res.json();
    const sel    = document.getElementById('trainRoute');
    if (!sel) return;
    const cur = sel.value;
    sel.innerHTML = '<option value="">Select Route</option>';
    routes.forEach(r => {
        sel.innerHTML += `<option value="${r.route_id}">${r.source} →
${r.destination}</option>`;
    });
    sel.value = cur;
}

function openAddTrain() {
    editingTrainId = null;
    document.getElementById('trainModalTitle').textContent = 'Add New
Train';
    document.getElementById('trainForm').reset();
    populateRouteDropdown();
    openModal('trainModal');
```

```
}

function openEditTrain(t) {
    editingTrainId = t.train_id;
    document.getElementById('trainModalTitle').textContent  =  'Edit
Train';
    populateRouteDropdown().then(() => {
        document.getElementById('trainRoute').value = t.route_id;
    });
    document.getElementById('trainName').value      = t.train_name;
    document.getElementById('trainNumber').value    = t.train_number;
    document.getElementById('trainType').value      = t.train_type;
    document.getElementById('trainDeparture').value                =
t.departure_time;
    document.getElementById('trainArrival').value   = t.arrival_time;
    document.getElementById('trainSeats').value     = t.total_seats;
    document.getElementById('trainDays').value                     =
t.days_of_operation;
    document.getElementById('trainStatus').value    = t.status;
    openModal('trainModal');
}

document.getElementById('trainForm').addEventListener('submit',
async (e) => {
    e.preventDefault();
    const data = {
        train_name:      document.getElementById('trainName').value,
        train_number:
document.getElementById('trainNumber').value,
        train_type:      document.getElementById('trainType').value,
        route_id:
document.getElementById('trainRoute').value,
        departure_time:
document.getElementById('trainDeparture').value,
        arrival_time:
document.getElementById('trainArrival').value,
        total_seats:
document.getElementById('trainSeats').value,
        days_of_operation:
document.getElementById('trainDays').value,
        status:         document.getElementById('trainStatus').value
    };
    if (editingTrainId) {
        data.train_id = editingTrainId;
        await  fetch(API  +  '?action=edit_train',  {  method:  'POST',
body: JSON.stringify(data) });
        showAlert('success', 'Train updated successfully!');
    } else {
        await fetch(API + '?action=add_train', { method: 'POST', body:
```

```
JSON.stringify(data) });
        showAlert('success', 'Train added successfully!');
    }
    closeModal('trainModal');
    loadTrains();
    loadStats();
});

async function deleteTrain(id) {
    if (!confirm('Delete this train?')) return;
    await fetch(API + '?action=delete_train&id=' + id);
    showAlert('success', 'Train deleted.');
    loadTrains();
    loadStats();
}

// ═══════════════════════════════════════════════
//  PASSENGERS
// ═══════════════════════════════════════════════
let editingPassengerId = null;

async function loadPassengers() {
    const res  = await fetch(API + '?table=passengers');
    const data = await res.json();
    let html = '';
    data.forEach(p => {
        html += `<tr>
            <td>${p.passenger_id}</td>
            <td>${p.first_name} ${p.last_name}</td>
            <td>${p.email}</td>
            <td>${p.phone}</td>
            <td>${p.gender}</td>
            <td>${p.id_proof_type}</td>
            <td><div class="action-buttons">
                <button      class="btn      btn-warning      btn-sm"
onclick='openEditPassenger(${JSON.stringify(p)})'>
                    <i class="fas fa-edit"></i> Edit
                </button>
                <button      class="btn      btn-danger      btn-sm"
onclick="deletePassenger(${p.passenger_id})">
                    <i class="fas fa-trash"></i> Delete
                </button>
            </div></td>
        </tr>`;
    });
    document.getElementById('passengersTable').innerHTML = html;
    populatePassengerDropdown(data);
}
```

```
function populatePassengerDropdown(passengers) {
    const sel = document.getElementById('ticketPassenger');
    if (!sel) return;
    const cur = sel.value;
    sel.innerHTML = '<option value="">Select Passenger</option>';
    passengers.forEach(p => {
        sel.innerHTML                   +=                   `<option
value="${p.passenger_id}">${p.first_name} ${p.last_name}</option>`;
    });
    sel.value = cur;
}

function openAddPassenger() {
    editingPassengerId = null;
    document.getElementById('passengerModalTitle').textContent     =
'Add New Passenger';
    document.getElementById('passengerForm').reset();
    openModal('passengerModal');
}

function openEditPassenger(p) {
    editingPassengerId = p.passenger_id;
    document.getElementById('passengerModalTitle').textContent     =
'Edit Passenger';
    document.getElementById('passengerFirstName').value            =
p.first_name;
    document.getElementById('passengerLastName').value             =
p.last_name;
    document.getElementById('passengerEmail').value       = p.email;
    document.getElementById('passengerPhone').value       = p.phone;
    document.getElementById('passengerDOB').value                  =
p.date_of_birth;
    document.getElementById('passengerGender').value     = p.gender;
    document.getElementById('passengerIDType').value               =
p.id_proof_type;
    document.getElementById('passengerIDNumber').value             =
p.id_proof_number;
    document.getElementById('passengerAddress').value              =
p.address;
    openModal('passengerModal');
}

document.getElementById('passengerForm').addEventListener('submit',
async (e) => {
    e.preventDefault();
    const data = {
        first_name:
document.getElementById('passengerFirstName').value,
        last_name:
```

```
document.getElementById('passengerLastName').value,
        email:
document.getElementById('passengerEmail').value,
        phone:
document.getElementById('passengerPhone').value,
        date_of_birth:
document.getElementById('passengerDOB').value,
        gender:
document.getElementById('passengerGender').value,
        id_proof_type:
document.getElementById('passengerIDType').value,
        id_proof_number:
document.getElementById('passengerIDNumber').value,
        address:
document.getElementById('passengerAddress').value
    };
    if (editingPassengerId) {
        data.passenger_id = editingPassengerId;
        await fetch(API + '?action=edit_passenger', { method: 'POST',
body: JSON.stringify(data) });
        showAlert('success', 'Passenger updated successfully!');
    } else {
        await fetch(API + '?action=add_passenger', { method: 'POST',
body: JSON.stringify(data) });
        showAlert('success', 'Passenger added successfully!');
    }
    closeModal('passengerModal');
    loadPassengers();
    loadStats();
});

async function deletePassenger(id) {
    if (!confirm('Delete this passenger? Their tickets will also be
deleted.')) return;
    await fetch(API + '?action=delete_passenger&id=' + id);
    showAlert('success', 'Passenger deleted.');
    loadPassengers();
    loadStats();
}

// ═══════════════════════════════════════════════
//  TICKETS
// ═══════════════════════════════════════════════
let editingTicketId = null;

async function loadTickets() {
    const res  = await fetch(API + '?table=tickets');
    const data = await res.json();
    let html = '';
```

```
    data.forEach(t => {
        html += `<tr>
            <td>${t.ticket_id}</td>
            <td><strong>${t.pnr_number}</strong></td>
            <td>${t.passenger_id}</td>
            <td>${t.train_id}</td>
            <td>${t.journey_date}</td>
            <td>${t.seat_number}</td>
            <td>${t.seat_class}</td>
            <td>₹${t.fare}</td>
            <td>${badgeHtml(t.ticket_status, statusBadge)}</td>
            <td><div class="action-buttons">
                <button      class="btn      btn-warning      btn-sm"
onclick='openEditTicket(${JSON.stringify(t)})'>
                    <i class="fas fa-edit"></i> Edit
                </button>
                ${t.ticket_status !== 'Cancelled' ? `
                <button      class="btn      btn-warning      btn-sm"
onclick="cancelTicket(${t.ticket_id})" style="background:#f59e0b">
                    <i class="fas fa-ban"></i> Cancel
                </button>` : ''}
                <button      class="btn      btn-danger      btn-sm"
onclick="deleteTicket(${t.ticket_id})">
                    <i class="fas fa-trash"></i> Delete
                </button>
            </div></td>
        </tr>`;
    });
    document.getElementById('ticketsTable').innerHTML = html;
    populateTicketDropdown(data);
}

function populateTicketDropdown(tickets) {
    const sel = document.getElementById('paymentTicket');
    if (!sel) return;
    const cur = sel.value;
    sel.innerHTML = '<option value="">Select Ticket</option>';
    tickets.forEach(t => {
        sel.innerHTML                  +=                  `<option
value="${t.ticket_id}">${t.pnr_number} — ₹${t.fare}</option>`;
    });
    sel.value = cur;
}

function openAddTicket() {
    editingTicketId = null;
    document.getElementById('ticketModalTitle').textContent  =  'Book
New Ticket';
    document.getElementById('ticketForm').reset();
```

```
    openModal('ticketModal');
}

function openEditTicket(t) {
    editingTicketId = t.ticket_id;
    document.getElementById('ticketModalTitle').textContent  =  'Edit
Ticket';
    document.getElementById('ticketPassenger').value               =
t.passenger_id;
    document.getElementById('ticketTrain').value       = t.train_id;
    document.getElementById('ticketJourneyDate').value             =
t.journey_date;
    document.getElementById('ticketClass').value      = t.seat_class;
    document.getElementById('ticketStatus').value                  =
t.ticket_status;
    document.getElementById('ticketFare').value        = t.fare;
    openModal('ticketModal');
}

document.getElementById('ticketForm').addEventListener('submit',
async (e) => {
    e.preventDefault();
    const data = {
        passenger_id:
document.getElementById('ticketPassenger').value,
        train_id:      document.getElementById('ticketTrain').value,
        journey_date:
document.getElementById('ticketJourneyDate').value,
        seat_class:    document.getElementById('ticketClass').value,
        ticket_status:
document.getElementById('ticketStatus').value,
        fare:           document.getElementById('ticketFare').value
    };
    if (editingTicketId) {
        data.ticket_id = editingTicketId;
        await  fetch(API  +  '?action=edit_ticket',  {  method:  'POST',
body: JSON.stringify(data) });
        showAlert('success', 'Ticket updated successfully!');
    } else {
        const  res      =  await  fetch(API  +  '?action=add_ticket',  {
method: 'POST', body: JSON.stringify(data) });
        const result = await res.json();
        showAlert('success',  `Ticket  booked!  PNR:  ${result.pnr},
Seat: ${result.seat}`);
    }
    closeModal('ticketModal');
    loadTickets();
    loadStats();
});
```

```
async function cancelTicket(id) {
    if (!confirm('Cancel this ticket?')) return;
    await fetch(API + '?action=cancel_ticket&id=' + id);
    showAlert('success', 'Ticket cancelled.');
    loadTickets();
}

async function deleteTicket(id) {
    if (!confirm('Permanently delete this ticket?')) return;
    await fetch(API + '?action=delete_ticket&id=' + id);
    showAlert('success', 'Ticket deleted.');
    loadTickets();
    loadStats();
}

// ═══════════════════════════════════════════════
//  PAYMENTS
// ═══════════════════════════════════════════════
let editingPaymentId = null;

async function loadPayments() {
    const res  = await fetch(API + '?table=payments');
    const data = await res.json();
    let html = '';
    data.forEach(p => {
        html += `<tr>
            <td>${p.payment_id}</td>
            <td>${p.ticket_id}</td>
            <td>₹${p.amount}</td>
            <td>${p.payment_method}</td>
            <td>${badgeHtml(p.payment_status, statusBadge)}</td>
            <td style="font-size:0.8rem">${p.transaction_id}</td>
            <td>${p.payment_date}</td>
            <td><div class="action-buttons">
                <button      class="btn      btn-warning      btn-sm"
onclick='openEditPayment(${JSON.stringify(p)})'>
                    <i class="fas fa-edit"></i> Edit
                </button>
                <button      class="btn      btn-danger      btn-sm"
onclick="deletePayment(${p.payment_id})">
                    <i class="fas fa-trash"></i> Delete
                </button>
            </div></td>
        </tr>`;
    });
    document.getElementById('paymentsTable').innerHTML = html;
}
```

```
function openAddPayment() {
    editingPaymentId = null;
    document.getElementById('paymentModalTitle').textContent  =  'Add
Payment';
    document.getElementById('paymentForm').reset();
    openModal('paymentModal');
}

function openEditPayment(p) {
    editingPaymentId = p.payment_id;
    document.getElementById('paymentModalTitle').textContent = 'Edit
Payment';
    document.getElementById('paymentTicket').value = p.ticket_id;
    document.getElementById('paymentAmount').value = p.amount;
    document.getElementById('paymentMethod').value                  =
p.payment_method;
    document.getElementById('paymentStatus').value                  =
p.payment_status;
    openModal('paymentModal');
}

document.getElementById('paymentForm').addEventListener('submit',
async (e) => {
    e.preventDefault();
    const data = {
        ticket_id:
document.getElementById('paymentTicket').value,
        amount:
document.getElementById('paymentAmount').value,
        payment_method:
document.getElementById('paymentMethod').value,
        payment_status:
document.getElementById('paymentStatus').value
    };
    if (editingPaymentId) {
        data.payment_id = editingPaymentId;
        await fetch(API + '?action=edit_payment', { method: 'POST',
body: JSON.stringify(data) });
        showAlert('success', 'Payment updated!');
    } else {
        const res     = await fetch(API + '?action=add_payment', {
method: 'POST', body: JSON.stringify(data) });
        const result = await res.json();
        showAlert('success', `Payment saved! TXN: ${result.txn}`);
    }
    closeModal('paymentModal');
    loadPayments();
    loadStats();
});
```

```
async function deletePayment(id) {
    if (!confirm('Delete this payment record?')) return;
    await fetch(API + '?action=delete_payment&id=' + id);
    showAlert('success', 'Payment deleted.');
    loadPayments();
    loadStats();
}

// ═══════════════════════════════════════════
//  SEAT AVAILABILITY
// ═══════════════════════════════════════════
let editingAvailabilityId = null;

async function loadAvailability() {
    const res  = await fetch(API + '?table=availability');
    const data = await res.json();
    let html = '';
    data.forEach(s => {
        const pct  = Math.round((s.available_seats / s.total_seats) *
100);
        const cls  = pct > 50 ? 'badge-success' : pct > 20 ? 'badge-
warning' : 'badge-danger';
        html += `<tr>
            <td>${s.availability_id}</td>
            <td>${s.train_id}</td>
            <td>${s.journey_date}</td>
            <td>${s.total_seats}</td>
            <td><span                                    class="badge
${cls}">${s.available_seats}</span></td>
            <td>${s.sleeper_available}</td>
            <td>${s.ac_3tier_available}</td>
            <td>${s.ac_2tier_available}</td>
            <td>${s.ac_1tier_available}</td>
            <td><div class="action-buttons">
                <button      class="btn      btn-warning      btn-sm"
onclick='openEditAvailability(${JSON.stringify(s)})'>
                    <i class="fas fa-edit"></i> Edit
                </button>
                <button      class="btn      btn-danger      btn-sm"
onclick="deleteAvailability(${s.availability_id})">
                    <i class="fas fa-trash"></i> Delete
                </button>
            </div></td>
        </tr>`;
    });
    document.getElementById('availabilityTable').innerHTML = html;
}
```

```
function openAddAvailability() {
    editingAvailabilityId = null;
    document.getElementById('availabilityModalTitle').textContent  =
'Add Seat Availability';
    document.getElementById('availabilityForm').reset();
    openModal('availabilityModal');
}

function openEditAvailability(s) {
    editingAvailabilityId = s.availability_id;
    document.getElementById('availabilityModalTitle').textContent  =
'Edit Seat Availability';
    document.getElementById('availabilityTrain').value             =
s.train_id;
    document.getElementById('availabilityDate').value              =
s.journey_date;
    document.getElementById('availabilityTotal').value             =
s.total_seats;
    document.getElementById('availabilityAvailable').value         =
s.available_seats;
    document.getElementById('availabilitySleeper').value           =
s.sleeper_available;
    document.getElementById('availability3AC').value               =
s.ac_3tier_available;
    document.getElementById('availability2AC').value               =
s.ac_2tier_available;
    document.getElementById('availability1AC').value               =
s.ac_1tier_available;
    openModal('availabilityModal');
}

document.getElementById('availabilityForm').addEventListener('submit
', async (e) => {
    e.preventDefault();
    const data = {
        train_id:
document.getElementById('availabilityTrain').value,
        journey_date:
document.getElementById('availabilityDate').value,
        total_seats:
document.getElementById('availabilityTotal').value,
        available_seats:
document.getElementById('availabilityAvailable').value,
        sleeper_available:
document.getElementById('availabilitySleeper').value,
        ac_3tier_available:
document.getElementById('availability3AC').value,
        ac_2tier_available:
document.getElementById('availability2AC').value,
```

```
        ac_1tier_available:
document.getElementById('availability1AC').value
    };
    if (editingAvailabilityId) {
        data.availability_id = editingAvailabilityId;
        await  fetch(API  +  '?action=edit_availability',  {  method:
'POST', body: JSON.stringify(data) });
        showAlert('success', 'Availability updated!');
    } else {
        await  fetch(API  +  '?action=add_availability',  {  method:
'POST', body: JSON.stringify(data) });
        showAlert('success', 'Seat availability added!');
    }
    closeModal('availabilityModal');
    loadAvailability();
});

async function deleteAvailability(id) {
    if (!confirm('Delete this availability record?')) return;
    await fetch(API + '?action=delete_availability&id=' + id);
    showAlert('success', 'Record deleted.');
    loadAvailability();
}

// ──────────────────────────────────────────────
// INIT  –  load everything on page load
// ──────────────────────────────────────────────
window.onload = () => {
    loadStats();
    loadRoutes();
    loadTrains();
    loadPassengers();
    loadTickets();
    loadPayments();
    loadAvailability();
};
```

PHP Source Files

config.php

```
<?php
$host = "localhost:3307";
$user = "root";
$password = "";
$database = "railway_reservation_system";
```

```
$conn = new mysqli($host, $user, $password, $database);

if ($conn->connect_error) {
    die("DB FAILED: " . $conn->connect_error);
}
?>
```

api.php

```
<?php
header("Content-Type: application/json");
header("Access-Control-Allow-Origin: *");
header("Access-Control-Allow-Methods:   GET,   POST,   PUT,   DELETE,
OPTIONS");
header("Access-Control-Allow-Headers: Content-Type");

if ($_SERVER['REQUEST_METHOD'] === 'OPTIONS') {
    http_response_code(200);
    exit;
}

$conn      =      new      mysqli("localhost",      "root",      "",
"railway_reservation_system", 3307);

if ($conn->connect_error) {
    die(json_encode(["error"  =>  "DB  Connection  Failed:  "  .  $conn-
>connect_error]));
}

$action = $_GET['action'] ?? '';
$table  = $_GET['table']  ?? '';
$id     = $_GET['id']     ?? '';

// ────────────────────────────────────────────
// GENERIC FETCH  →  GET ?table=ROUTE  etc.
// ────────────────────────────────────────────
if ($table) {
    $allowed                                                          =
['routes','trains','passengers','availability','tickets','payments']
;
    if (!in_array(strtolower($table), $allowed)) {
        echo json_encode(["error" => "Invalid table"]);
        exit;
    }
    $t      = strtolower($table);
    $result = $conn->query("SELECT * FROM $t");
    $data   = [];
    while ($row = $result->fetch_assoc()) {
```

```
        $data[] = $row;
    }
    echo json_encode($data);
    exit;
}

// Read input body once
$input = json_decode(file_get_contents("php://input"), true) ?? [];

// ═══════════════════════════════════════════════
//  ROUTES
// ═══════════════════════════════════════════════

if ($action === "add_route") {
    $stmt = $conn->prepare(
        "INSERT  INTO  routes  (source,  destination,  distance_km,
duration_hours)
         VALUES (?, ?, ?, ?)"
    );
    $stmt->bind_param("ssdd",
        $input['source'], $input['destination'],
        $input['distance_km'], $input['duration_hours']
    );
    $stmt->execute();
    echo json_encode(["success" => true, "id" => $conn->insert_id]);
    exit;
}

if ($action === "edit_route") {
    $stmt = $conn->prepare(
        "UPDATE  routes  SET  source=?,  destination=?,  distance_km=?,
duration_hours=?
         WHERE route_id=?"
    );
    $stmt->bind_param("ssddi",
        $input['source'], $input['destination'],
        $input['distance_km'], $input['duration_hours'],
        $input['route_id']
    );
    $stmt->execute();
    echo json_encode(["success" => true]);
    exit;
}

if ($action === "delete_route") {
    $conn->query("DELETE FROM routes WHERE route_id=" . (int)$id);
    echo json_encode(["success" => true]);
    exit;
}
```

```
// ═══════════════════════════════════════════
//  TRAINS
// ═══════════════════════════════════════════

if ($action === "add_train") {
    $stmt = $conn->prepare(
        "INSERT INTO trains (train_name, train_number, train_type,
total_seats,
                            route_id, departure_time, arrival_time,
                            days_of_operation, status)
         VALUES (?, ?, ?, ?, ?, ?, ?, ?, ?)"
    );
    $stmt->bind_param("sssiiisss",
        $input['train_name'],                $input['train_number'],
$input['train_type'],
        $input['total_seats'], $input['route_id'],
        $input['departure_time'], $input['arrival_time'],
        $input['days_of_operation'], $input['status']
    );
    $stmt->execute();
    echo json_encode(["success" => true, "id" => $conn->insert_id]);
    exit;
}

if ($action === "edit_train") {
    $stmt = $conn->prepare(
        "UPDATE    trains    SET    train_name=?,    train_number=?,
train_type=?,
         total_seats=?, route_id=?, departure_time=?, arrival_time=?,
         days_of_operation=?, status=?
         WHERE train_id=?"
    );
    $stmt->bind_param("sssiiisssi",
        $input['train_name'],                $input['train_number'],
$input['train_type'],
        $input['total_seats'], $input['route_id'],
        $input['departure_time'], $input['arrival_time'],
        $input['days_of_operation'], $input['status'],
        $input['train_id']
    );
    $stmt->execute();
    echo json_encode(["success" => true]);
    exit;
}

if ($action === "delete_train") {
    $conn->query("DELETE FROM trains WHERE train_id=" . (int)$id);
    echo json_encode(["success" => true]);
```

```
    exit;
}

// ==========================================
//  PASSENGERS
// ==========================================

if ($action === "add_passenger") {
    $stmt = $conn->prepare(
        "INSERT INTO passengers (first_name, last_name, email, phone,
                                 date_of_birth, gender, address,
                                 id_proof_type, id_proof_number)
         VALUES (?, ?, ?, ?, ?, ?, ?, ?, ?)"
    );
    $stmt->bind_param("sssssssss",
        $input['first_name'], $input['last_name'],
        $input['email'], $input['phone'],
        $input['date_of_birth'], $input['gender'],
        $input['address'], $input['id_proof_type'],
        $input['id_proof_number']
    );
    $stmt->execute();
    echo json_encode(["success" => true, "id" => $conn->insert_id]);
    exit;
}

if ($action === "edit_passenger") {
    $stmt = $conn->prepare(
        "UPDATE passengers SET first_name=?, last_name=?, email=?,
phone=?,
         date_of_birth=?,  gender=?,  address=?,  id_proof_type=?,
id_proof_number=?
         WHERE passenger_id=?"
    );
    $stmt->bind_param("ssssssssi",
        $input['first_name'], $input['last_name'],
        $input['email'], $input['phone'],
        $input['date_of_birth'], $input['gender'],
        $input['address'], $input['id_proof_type'],
        $input['id_proof_number'], $input['passenger_id']
    );
    $stmt->execute();
    echo json_encode(["success" => true]);
    exit;
}

if ($action === "delete_passenger") {
    $conn->query("DELETE FROM passengers WHERE passenger_id=" .
(int)$id);
```

```
    echo json_encode(["success" => true]);
    exit;
}

// ═══════════════════════════════════════════
//  TICKETS
// ═══════════════════════════════════════════

if ($action === "add_ticket") {
    // auto-generate PNR
    $pnr = 'PNR' . str_pad(rand(0, 9999999), 7, '0', STR_PAD_LEFT);
    $seatPrefix = ['Sleeper'=>'S','3AC'=>'B','2AC'=>'A','1AC'=>'F'];
    $prefix = $seatPrefix[$input['seat_class']] ?? 'S';
    $seatNum = $prefix . rand(1,10) . '-' . rand(1,72);

    $stmt = $conn->prepare(
        "INSERT  INTO  tickets  (pnr_number,  passenger_id,  train_id,
journey_date,
                             seat_number, seat_class, ticket_status,
fare)
         VALUES (?, ?, ?, ?, ?, ?, ?, ?)"
    );
    $stmt->bind_param("siissssd",
        $pnr, $input['passenger_id'], $input['train_id'],
        $input['journey_date'], $seatNum,
        $input['seat_class'],               $input['ticket_status'],
$input['fare']
    );
    $stmt->execute();
    echo  json_encode(["success"  =>  true,  "id"  =>  $conn->insert_id,
"pnr" => $pnr, "seat" => $seatNum]);
    exit;
}

if ($action === "edit_ticket") {
    $stmt = $conn->prepare(
        "UPDATE    tickets    SET    passenger_id=?,    train_id=?,
journey_date=?,
         seat_class=?, ticket_status=?, fare=?
         WHERE ticket_id=?"
    );
    $stmt->bind_param("iissdi",  // removed wrong 'ssd' combo – fixed
below
        $input['passenger_id'], $input['train_id'],
        $input['journey_date'], $input['seat_class'],
        $input['ticket_status'], $input['fare'],
        $input['ticket_id']
    );
    $stmt->execute();
```

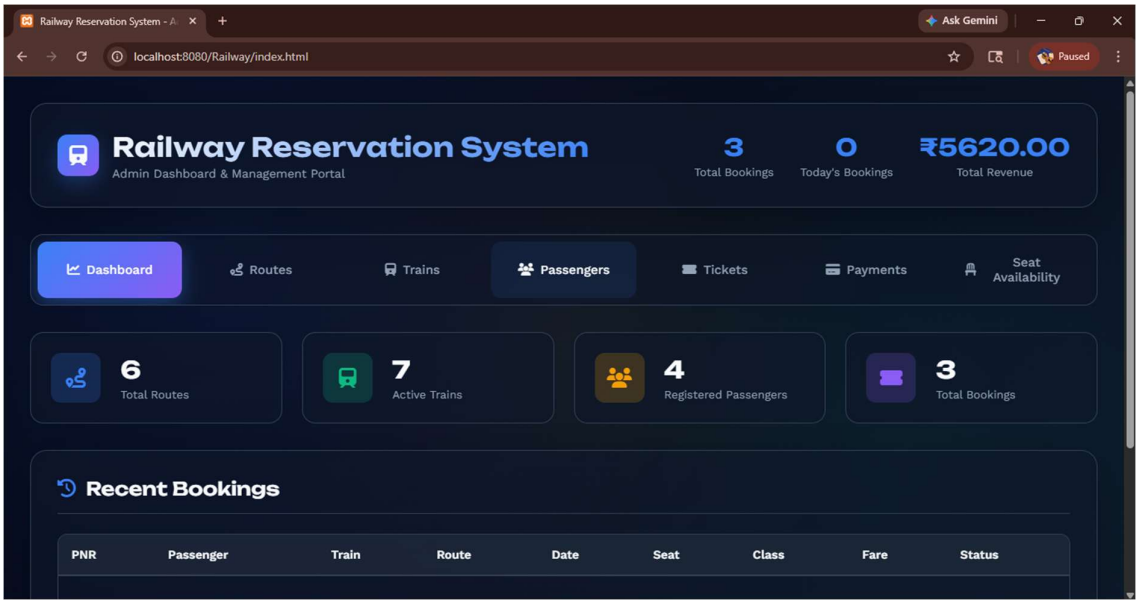
```
    echo json_encode(["success" => true]);
    exit;
}

if ($action === "cancel_ticket") {
    $conn->query("UPDATE tickets SET ticket_status='Cancelled' WHERE
ticket_id=" . (int)$id);
    echo json_encode(["success" => true]);
    exit;
}

if ($action === "delete_ticket") {
    $conn->query("DELETE FROM tickets WHERE ticket_id=" . (int)$id);
    echo json_encode(["success" => true]);
    exit;
}

// ═══════════════════════════════════════════
//  PAYMENTS
// ═══════════════════════════════════════════

if ($action === "add_payment") {
    $txn = 'TXN' . time() . rand(100, 999);
    $stmt = $conn->prepare(
        "INSERT INTO payments (ticket_id, amount, payment_method,
payment_status, transaction_id)
         VALUES (?, ?, ?, ?, ?)"
    );
    $stmt->bind_param("idsss",
        $input['ticket_id'], $input['amount'],
        $input['payment_method'], $input['payment_status'], $txn
    );
    $stmt->execute();
    echo json_encode(["success" => true, "id" => $conn->insert_id,
"txn" => $txn]);
    exit;
}

if ($action === "edit_payment") {
    $stmt = $conn->prepare(
        "UPDATE payments SET amount=?, payment_method=?,
payment_status=?
         WHERE payment_id=?"
    );
    $stmt->bind_param("dssi",
        $input['amount'], $input['payment_method'],
        $input['payment_status'], $input['payment_id']
    );
    $stmt->execute();
```

```
    echo json_encode(["success" => true]);
    exit;
}

if ($action === "delete_payment") {
    $conn->query("DELETE   FROM   payments   WHERE   payment_id="   .
(int)$id);
    echo json_encode(["success" => true]);
    exit;
}

// ═══════════════════════════════════════════
//  SEAT AVAILABILITY
// ═══════════════════════════════════════════

if ($action === "add_availability") {
    $stmt = $conn->prepare(
        "INSERT   INTO   availability   (train_id,   journey_date,
total_seats,
         available_seats, sleeper_available, ac_3tier_available,
         ac_2tier_available, ac_1tier_available)
         VALUES (?, ?, ?, ?, ?, ?, ?, ?)"
    );
    $stmt->bind_param("isiiiiii",
        $input['train_id'], $input['journey_date'],
        $input['total_seats'], $input['available_seats'],
        $input['sleeper_available'], $input['ac_3tier_available'],
        $input['ac_2tier_available'], $input['ac_1tier_available']
    );
    $stmt->execute();
    echo json_encode(["success" => true, "id" => $conn->insert_id]);
    exit;
}

if ($action === "edit_availability") {
    $stmt = $conn->prepare(
        "UPDATE availability SET train_id=?, journey_date=?,
         total_seats=?, available_seats=?, sleeper_available=?,
         ac_3tier_available=?,                 ac_2tier_available=?,
ac_1tier_available=?
         WHERE availability_id=?"
    );
    $stmt->bind_param("isiiiiii" . "i",
        $input['train_id'], $input['journey_date'],
        $input['total_seats'], $input['available_seats'],
        $input['sleeper_available'], $input['ac_3tier_available'],
        $input['ac_2tier_available'], $input['ac_1tier_available'],
        $input['availability_id']
    );
```

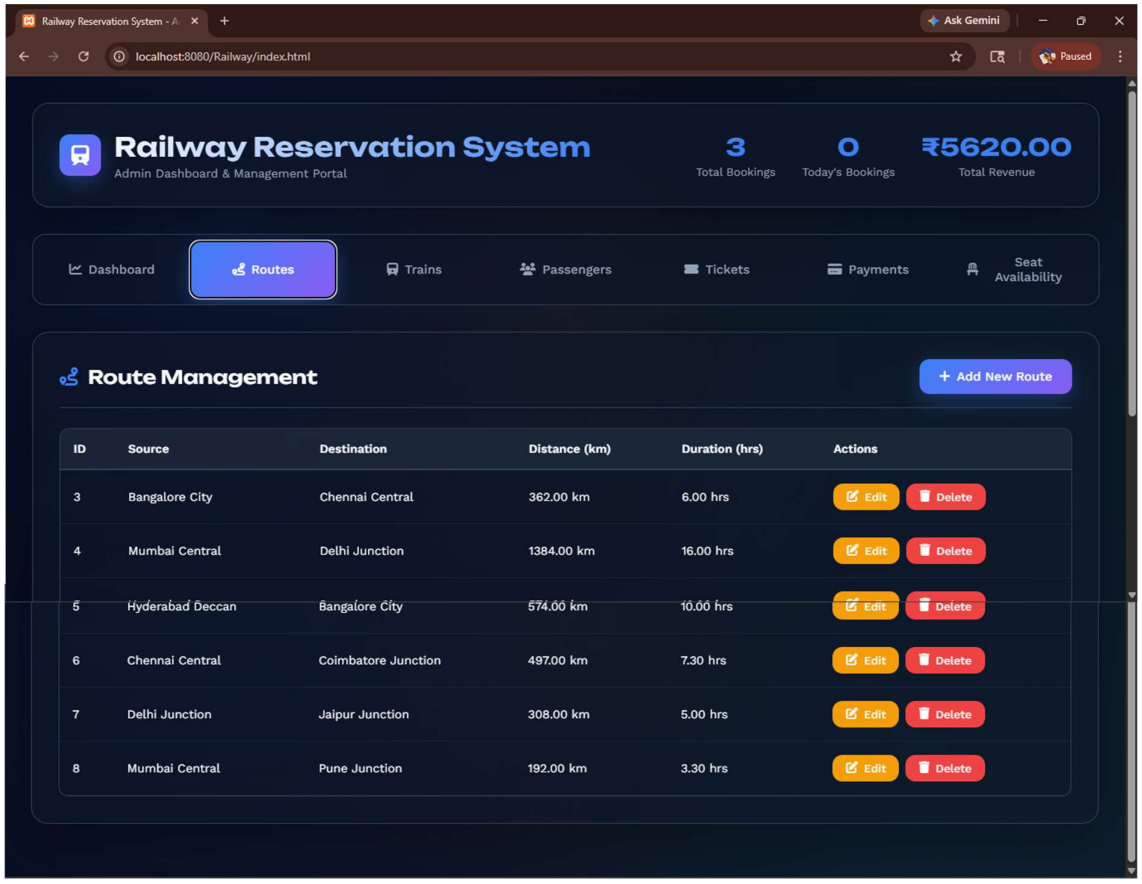
```
    $stmt->execute();
    echo json_encode(["success" => true]);
    exit;
}

if ($action === "delete_availability") {
    $conn->query("DELETE FROM availability WHERE availability_id=" .
(int)$id);
    echo json_encode(["success" => true]);
    exit;
}

// ═══════════════════════════════════════════════
//  DASHBOARD STATS
// ═══════════════════════════════════════════════

if ($action === "stats") {
    $stats = [];
    $stats['total_routes']     = $conn->query("SELECT COUNT(*) c FROM
routes")->fetch_assoc()['c'];
    $stats['total_trains']     = $conn->query("SELECT COUNT(*) c FROM
trains")->fetch_assoc()['c'];
    $stats['total_passengers'] = $conn->query("SELECT COUNT(*) c FROM
passengers")->fetch_assoc()['c'];
    $stats['total_tickets']    = $conn->query("SELECT COUNT(*) c FROM
tickets")->fetch_assoc()['c'];
    $stats['today_tickets']    = $conn->query(
        "SELECT      COUNT(*)      c      FROM      tickets      WHERE
DATE(booking_date)=CURDATE()"
    )->fetch_assoc()['c'];
    $rev = $conn->query(
        "SELECT   COALESCE(SUM(amount),0)   r   FROM   payments   WHERE
payment_status='Success'"
    )->fetch_assoc()['r'];
    $stats['total_revenue'] = number_format((float)$rev, 2, '.', '');
    echo json_encode($stats);
    exit;
}
echo json_encode(["error" => "Unknown action"]);
```

OUTPUT SCREENSHOTS

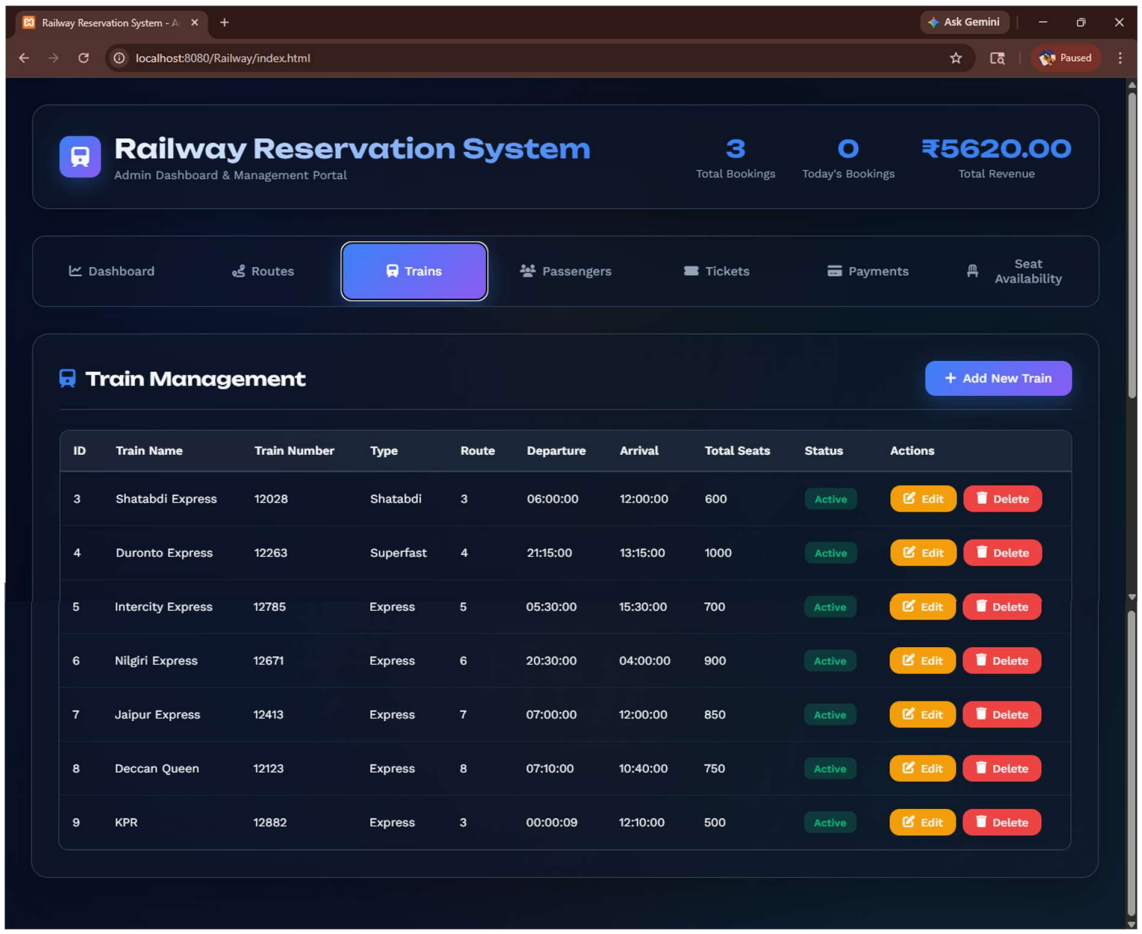
Dashboard:



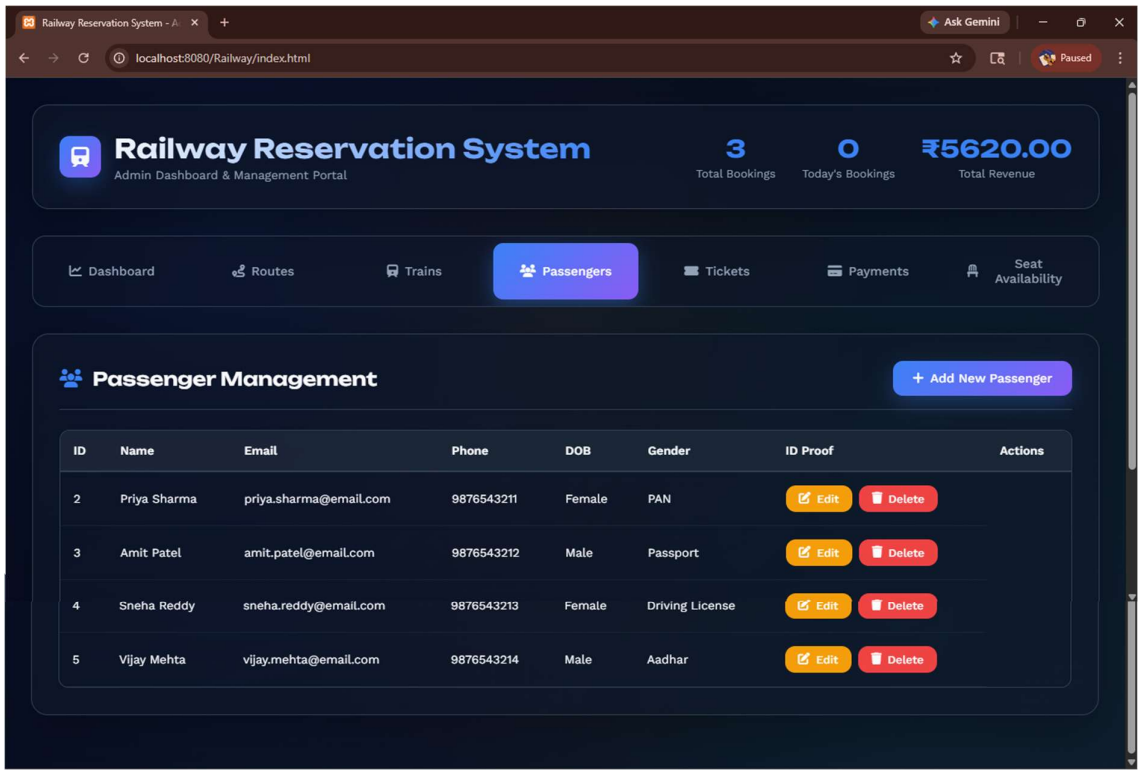
Routes:



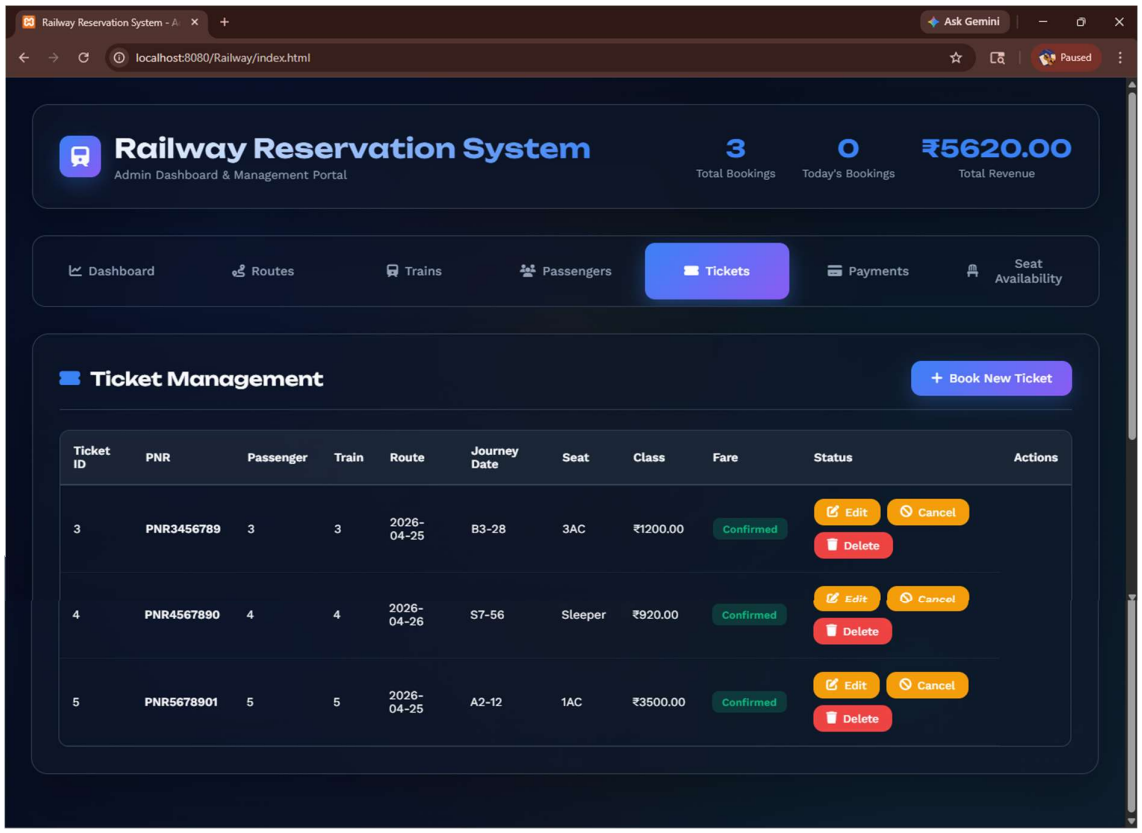
Trains:



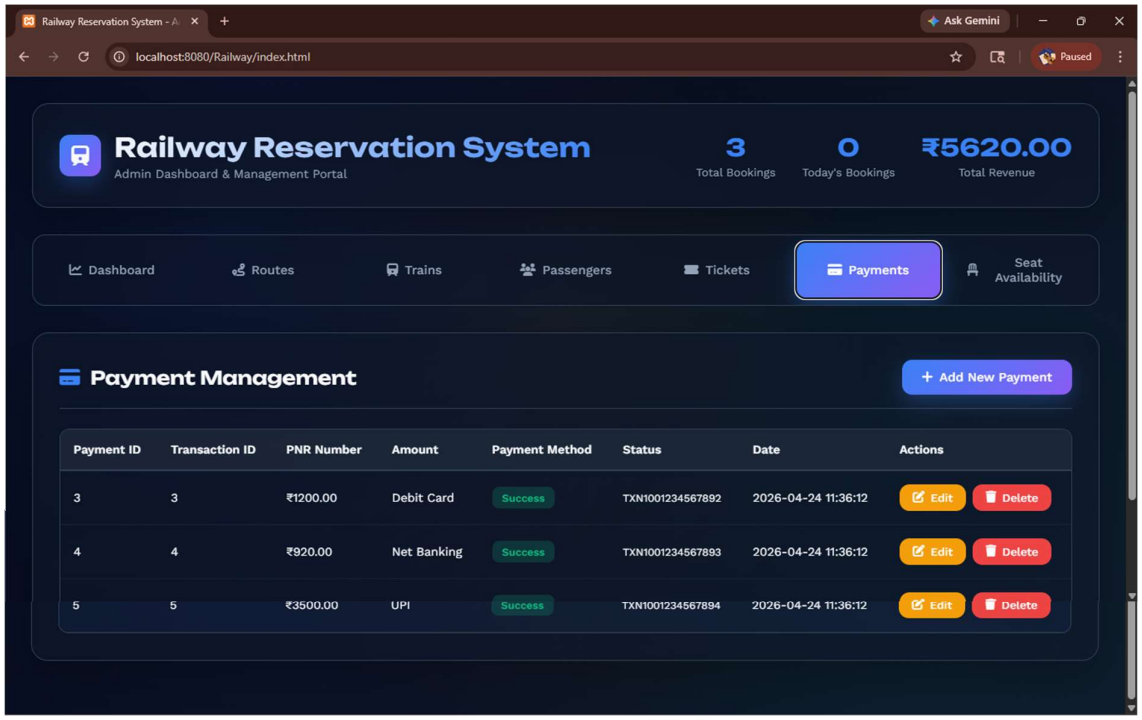
Passengers:



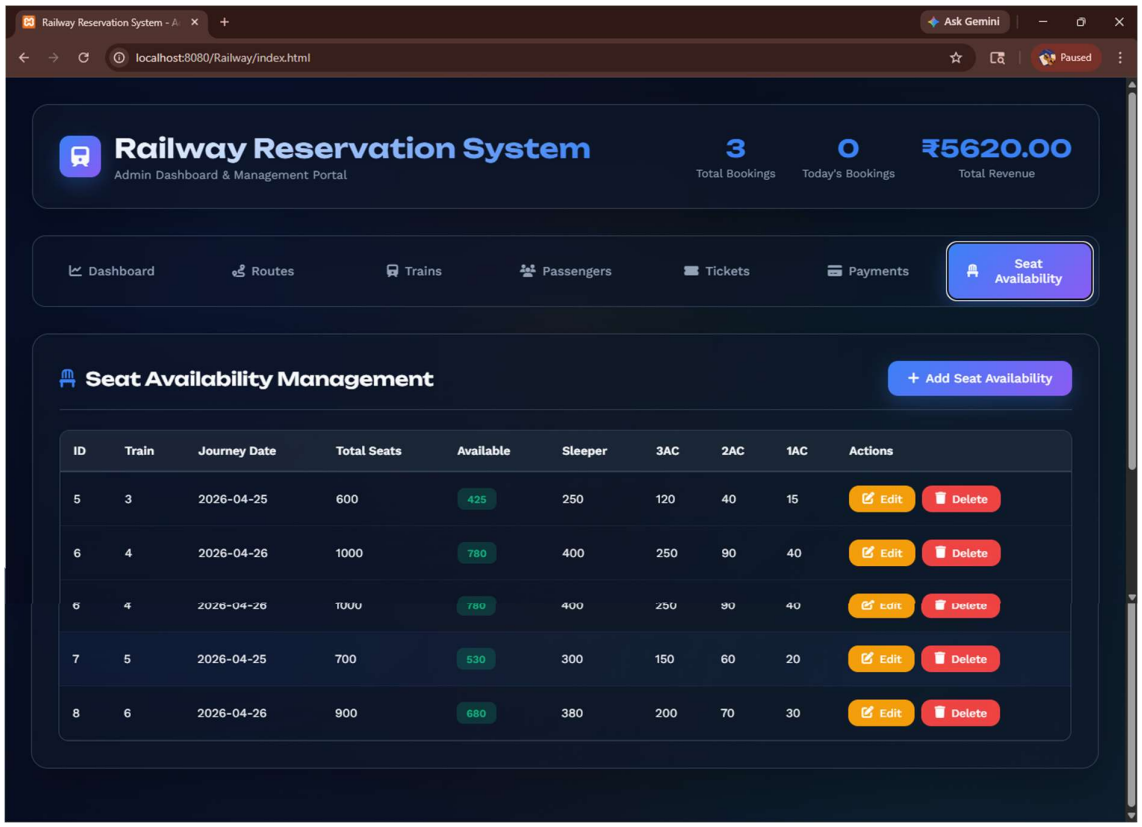
Tickets:



Payments:

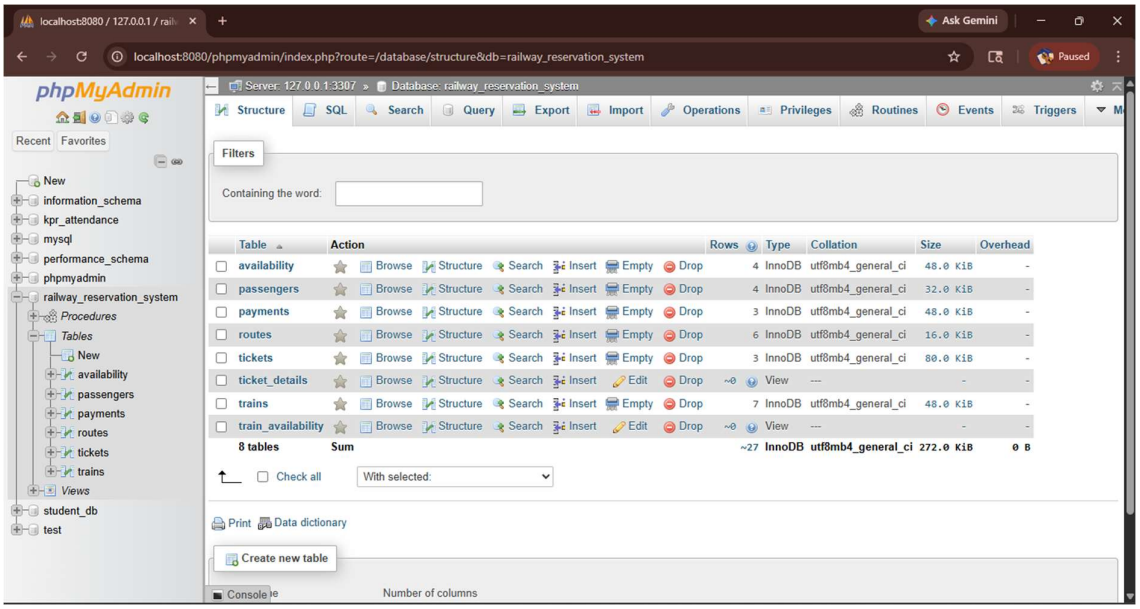


Seat Availability:



BACKEND DATABASE SCREENSHOTS

Tables in Database:



Availability:

availability_id	train_id	journey_date	total_seats	available_seats	sleeper_available	ac_3tier_available	ac_2tier_available	ac_1tier_available	updated_at
5	3	2026-04-25	600	425	250	120	40	15	2026-04-24 11:36:12
6	4	2026-04-26	1000	780	400	250	90	40	2026-04-24 11:36:12
7	5	2026-04-25	700	530	300	150	60	20	2026-04-24 11:36:12
8	6	2026-04-26	900	680	380	200	70	30	2026-04-24 11:36:12

Passengers:

passenger_id	first_name	last_name	email	phone	date_of_birth	gender	address	id_proof_type	id_proof_number	created_at
2	Priya	Sharma	priya.sharma@email.com	9876543211	1990-07-22	Female	456 Park Street, Mumbai	PAN	ABCDE1234F	2026-04-24 11:36:12
3	Amit	Patel	amit.patel@email.com	9876543212	1988-11-10	Male	789 Brigade Road, Bangalore	Passport	M1234567	2026-04-24 11:36:12
4	Sneha	Reddy	sneha.reddy@email.com	9876543213	1992-05-18	Female	321 Connaught Place, Delhi	Driving License	DL-1234567890	2026-04-24 11:36:12
5	Vijay	Mehta	vijay.mehta@email.com	9876543214	1987-09-25	Male	654 MG Road, Pune	Aadhar	9876-5432-1098	2026-04-24 11:36:12

Payments:

payment_id	ticket_id	amount	payment_method	payment_status	transaction_id	payment_date
3	3	1200.00	Debit Card	Success	TXN1001234567892	2026-04-24 11:36:12
4	4	920.00	Net Banking	Success	TXN1001234567893	2026-04-24 11:36:12
5	5	3500.00	UPI	Success	TXN1001234567894	2026-04-24 11:36:12

Tickets:

ticket_id	pnr_number	passenger_id	train_id	journey_date	seat_number	seat_class	booking_date	ticket_status	fare
3	PNR3456789	3	3	2026-04-25	B3-28	3AC	2026-04-24 11:36:12	Confirmed	1200.00
4	PNR4567890	4	4	2026-04-26	S7-56	Sleeper	2026-04-24 11:36:12	Confirmed	920.00
5	PNR5678901	5	5	2026-04-25	A2-12	1AC	2026-04-24 11:36:12	Confirmed	3500.00

Routes:

route_id	source	destination	distance_km	duration_hours	created_at
3	Bangalore City	Chennai Central	362.00	6.00	2026-04-24 11:36:12
4	Mumbai Central	Delhi Junction	1384.00	16.00	2026-04-24 11:36:12
5	Hyderabad Deccan	Bangalore City	574.00	10.00	2026-04-24 11:36:12
6	Chennai Central	Coimbatore Junction	497.00	7.30	2026-04-24 11:36:12
7	Delhi Junction	Jaipur Junction	308.00	5.00	2026-04-24 11:36:12
8	Mumbai Central	Pune Junction	192.00	3.30	2026-04-24 11:36:12

Trains:

train_id	train_name	train_number	train_type	total_seats	route_id	departure_time	arrival_time	days_of_operation	status	created_at
3	Shatabdi Express	12028	Shatabdi	600	3	06:00:00	12:00:00	All Days	Active	2026-04-24 11:36:12
4	Duronto Express	12263	Superfast	1000	4	21:15:00	13:15:00	All Days	Active	2026-04-24 11:36:12
5	Intercity Express	12785	Express	700	5	05:30:00	15:30:00	All Days	Active	2026-04-24 11:36:12
6	Nilgiri Express	12671	Express	900	6	20:30:00	04:00:00	All Days	Active	2026-04-24 11:36:12
7	Jaipur Express	12413	Express	850	7	07:00:00	12:00:00	All Days	Active	2026-04-24 11:36:12
8	Deccan Queen	12123	Express	750	8	07:10:00	10:40:00	All Days	Active	2026-04-24 11:36:12
9	KPR	12882	Express	500	3	00:00:09	12:10:00	All Days	Active	2026-04-24 14:54:26

RESULT

The Railway Reservation System was successfully designed and implemented by establishing proper connectivity between the frontend and backend using a local server environment. The system integrates **HTML, CSS, and JavaScript** for the frontend, PHP for server-side processing, and **MySQL for database management through XAMPP.**

The connectivity between the frontend and backend was successfully achieved using XAMPP, where Apache acted as the web server and MySQL as the database server.

Overall, the system was tested and verified for correctness, completeness, and efficiency. All modules functioned as expected, and the integration between the frontend, backend, and database was smooth and reliable.

CONCLUSION

Thus, the Railway Reservation System was successfully developed with proper database design, efficient connectivity setup, and functional user interface. The system demonstrates effective use of web technologies and database management concepts to perform real-time railway booking operations accurately and efficiently. The system provides a structured and efficient approach to managing railway reservations by integrating multiple modules such as booking, payment processing, and seat availability tracking. Each module works in coordination with the database, ensuring that the overall system performs seamlessly without data inconsistency.

In conclusion, the Railway Reservation System demonstrates the practical application of web development and database management concepts. It serves as a strong foundation for building more complex and scalable applications, highlighting the importance of proper connectivity, data handling, and system design.